NRC Research Press

ARTICLE

Landscape, colonization, and life history: their effects on genetic diversity in four sympatric species inhabiting a dendritic system

Daniel E. Ruzzante, Gregory R. McCracken, Sarah J. Salisbury, Hilary T. Brewis, Donald Keefe, Oscar E. Gaggiotti, and Robert Perry

Abstract: To what degree are patterns of genetic structure in fragmented systems the result of contemporary landscape versus history? We examined the distribution of genetic diversity as a function of colonization history and contemporary landscape in four fish species inhabiting a hierarchically fragmented, unaltered system, the Kogaluk drainage (Labrador): lake trout (*Salvelinus namaycush*), longnose sucker (*Catostomus catostomus*), round whitefish (*Prosopium cylindraceum*), and lake chub (*Couesius plumbeus*). The footprint of colonization history was still observable in the three species where this issue was examined regardless of the generations since their arrival. Approximate Bayesian computation (ABC) analyses suggest colonization took place from the southwest. The species exhibit similar diversity patterns despite different $\hat{N}_e$ values and generation intervals. Contemporary gene flow was largely negligible except for gene flow from a centrally located lake. These results suggest landscape has driven colonization history, which still has influence on genetic structuring. The species are widespread. Understanding how they behave in the pristine Kogaluk provides a baseline against which to evaluate how other anthropogenically perturbed systems are performing. Improved understanding of historical and contemporary processes is required to fully explain diversity patterns in complex metapopulations.

Résumé : Dans quelle mesure les motifs de structure génétique dans les systèmes fragmentés sont-ils le résultat du paysage actuel plutôt que de l'évolution passée? Nous avons examiné la répartition de la diversité génétique en fonction de l'historique de colonisation et du paysage actuel pour quatre espèces de poissons occupant un système non altéré caractérisé par une fragmentation hiérarchique, le bassin versant de Kogaluk (Labrador), soit le touladi (*Salvelinus namaycush*), le meunier rouge (*Catostomus catostomus*), le ménomini rond (*Prosopium cylindraceum*) et le méné de lac (*Couesius plumbeus*). L'empreinte de l'historique de colonisation est toujours perceptible chez les espèces pour lesquelles cet aspect a été examiné, peu importe le nombre de générations depuis leur arrivée. Des analyses ABC (« approximate Bayesian computation ») semblent indiquer que la colonisation s'est effectuée à partir du sud-ouest. Les espèces présentent des motifs de diversité semblables malgré des valeurs des $\hat{N}_e$ et intervalles entre générations différents. Le flux génétique actuel est généralement négligeable, sauf en ce qui concerne le flux génétique à partir d'un lac à situation centrale. Ces résultats donnent à penser que le paysage a modulé l'historique de colonisation, qui exerce toujours une influence sur la structuration génétique. Les espèces sont ubiquistes. La compréhension de leur comportement dans le bassin versant intact de Kogaluk fournit une référence par rapport à laquelle évaluer la performance de systèmes perturbés par l'activité humaine. Une meilleure compréhension des processus passés et actuels est nécessaire pour expliquer en détail les motifs de diversité dans les métapopulations complexes. [Traduit par la Rédaction]

Introduction

To what degree are patterns of observed genetic diversity in complex metapopulation systems the result of contemporary versus historical processes? This is a fundamental question in current conservation and landscape genetic studies; an understanding of their relative roles can assist in the assessment of future population responses to climate and (or) habitat changes (Manel et al. 2003; Storfer et al. 2010; Manel and Holderegger 2013). The question is, however, not easily answered since it requires the examination of processes operating over different time scales. While the genetic structure of any taxon may in principle be dictated by the interaction of historical and contemporary processes, this is particularly likely in those metapopulations that have failed to reach migration–drift equilibrium (e.g., those exposed to the Quaternary glacial cycles; Hewitt 2000; Ruzzante et al. 2008; Fraser et al. 2012).

In a contemporary setting, habitat fragmentation is generally thought to lead to the loss of genetic diversity. Yet this is not always true; spatial configuration of the habitat fragments also matters. For instance, systems that are spatially fragmented in a hierarchical, dendritic fashion have been shown, at least in theory, to exhibit higher genetic diversity than panmictic systems of equal total size (Morrissey and de Kerckhove 2009). Asymmetric gene flow occurring in these types of systems can significantly

Received 23 October 2018. Accepted 2 April 2019.

D.E. Ruzzante, G.R. McCracken, S.J. Salisbury, and H.T. Brewis.* Dept. of Biology, Dalhousie University, 1355 Oxford Street, Halifax, NS B3H 4R2, Canada.
D. Keefe and R. Perry. Newfoundland and Labrador Department of Fisheries and Land Resources, 117 Riverside Drive, Corner Brook, NL A2H 7S1, Canada.
O.E. Gaggiotti. Scottish Oceans Institute, East Sands, Univ. of St Andrews, St Andrews, Fife, KY16 8LB, UK.
Corresponding author: Daniel E. Ruzzante (email: daniel.ruzzante@dal.ca).
*Present address: BC Children's Hospital Research Institute, 950 West 28th Avenue, Vancouver, BC V5Z 4H4, Canada.
Copyright remains with the author(s) or their institution(s). Permission for reuse (free in most cases) can be obtained from RightsLink.

influence the distribution of genetic variation, with headwater populations typically exhibiting lower genetic diversity (i.e., lower heterozygosity, allelic richness) and higher genetic differentiation than downstream populations (Caldera and Bolnick 2008; Morrissey and de Kerckhove 2009; Junker et al. 2012). Downstream populations are thus expected to exhibit higher effective population sizes (N_e) than headwater populations (Morrissey and de Kerckhove 2009; Gomez-Uchida et al. 2009, 2013).

Extrapolating from modeling studies to the natural world, however, is not straightforward. Theoretical studies examining spatially fragmented systems generally focus on standardized ideal systems where migration follows an island, a one- or two-dimension stepping stone, a circular, or, as above, a perfectly dendritic model (but see Tufto and Hindar 2003; Hössjer et al. 2014, 2015). Most importantly though, modeling studies generally assume migration–drift equilibrium. Such efforts are undoubtedly useful for holistic understanding and serve as predictive tools under some circumstances, but the majority of natural systems are not easily classifiable into any of these discrete categories; they are vastly more complex, making predictions of their behavior under most situations more difficult. Regardless, model validation with empirical data are likely to be useful for an improved understanding of the natural world (see, e.g., Campbell Grant et al. 2007; Perkin and Gido 2012).

Natural systems are unlikely to be in migration–drift equilibrium, a factor that is expected to bring the footprint of colonization history to the forefront at the expense of the influence of contemporary landscape features and the spatial arrangement of populations. Such a footprint of colonization history is in fact expected to be particularly strong in recently colonized systems, with genetic diversity potentially being a function of successive founding events (e.g., Vera-Escalona et al. 2015, 2018; Salisbury et al. 2016) and the number of generations elapsed since first arrival regardless of the spatial arrangement of populations.

In the present study, we tested the hypotheses that colonization history could explain observed patterns of structure and connectivity and that this relationship would be a function of the number of generations elapsed since deglaciation (i.e., an inverse function of generation interval). We examined the relative influence of colonization history and landscape structure among species differing in life history traits and coexisting sympatrically in a spatially fragmented and hierarchically structured freshwater system in northern Labrador, the Kogaluk River system. Although hierarchical in nature, this system departs from the standard dendritic scenario addressed by most previous theoretical studies. First, local populations are restricted to the lakes (nodes in typical dendritic models) as opposed to dendritic systems where they are also present in the rivers (branches) connecting the lakes (Campbell Grant et al. 2007; Labonne et al. 2008; Perkin and Gido 2012). Second, although gene flow is asymmetric, in some cases differences in elevation among lakes and the presence of ice during a large part of the year reduce the probability of sustained gene flow, which can lead to a much slower progression towards migration–drift equilibrium.

Genetic diversity was examined in four species: lake trout (*Salvelinus namaycush*), longnose sucker (*Catostomus catostomus*), round whitefish (*Prosopium cylindraceum*), and lake chub (*Couesius plumbeus*). These species are widespread and inhabit similar landscapes throughout Canada and have close relatives in many temperate regions of the Northern Hemisphere. Gaining an understanding of how this particular system behaves under pristine conditions without anthropogenic interference can provide a baseline against which it is possible to evaluate how other systems undergoing anthropogenic perturbations are performing. In addition, the species differ in some key life history traits, chiefly among them being generation time (T), with $T_{\text{lake trout}} > T_{\text{longnose sucker}} > T_{\text{round whitefish}} > T_{\text{lake chub}}$, suggesting they may be at different stages of a progression to migration–drift equilibrium. Colonization history was examined in the first three species. We thus tested for the interactive effects of landscape structure and colonization history in species that differ in life history traits. We sought to identify the relative roles of colonization history and current environmental barriers on the metapopulation genetic structure of four coexisting fish species potentially differing in the progression to equilibrium conditions. Explicitly, we expect species with shorter generation times to achieve migration–drift equilibrium in a shorter time period. Observed patterns of population structure in such species would be more influenced by current landscape features than by colonization history when compared with species with longer generation times.

Methods

Study site and sample collection

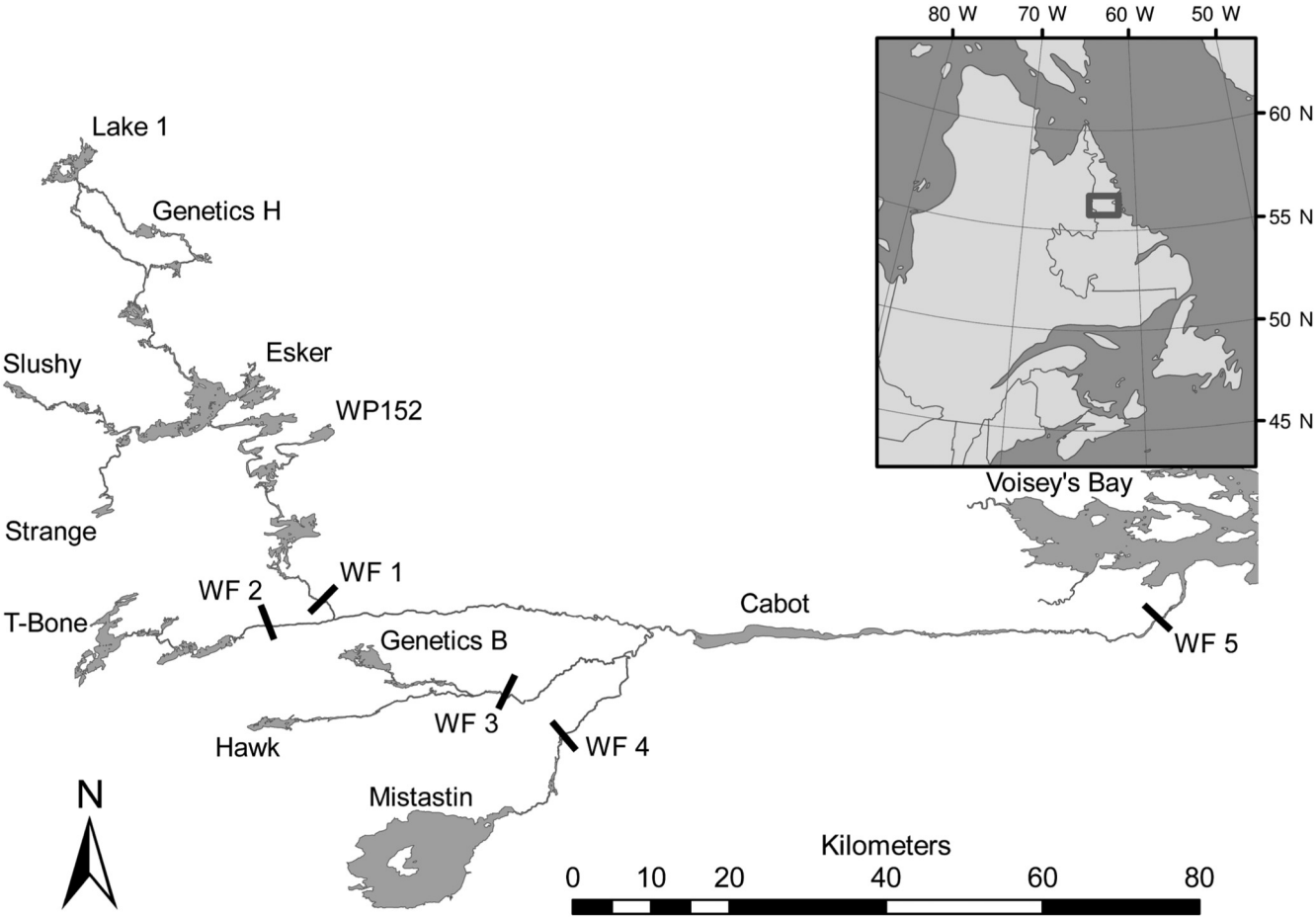
The Kogaluk River drainage comprises a number of relatively shallow lakes on the barren grounds of northern Labrador. The lakes are hierarchically connected through shallow streams that drain from the north, west, and south into Cabot Lake, a deep fjord lake. Cabot Lake, in turn, empties into the Atlantic Ocean through the Kogaluk River (Fig. 1). The lakes range in elevation from 525 m above sea level (a.s.l.) for the highest elevation to 60 m a.s.l. for Cabot lake (Table 1; Fig. 1). There are five waterfalls in the system, all of which prevent upstream migration. The furthest downstream waterfall is located just 9 km upstream from the river's mouth (Fig. 1; WF5; Anderson 1985), effectively rendering the fish populations in the entire system landlocked (Anderson 1985). Owing to the drainage's northern geographical location and distance from settled areas, the system is largely free of anthropogenic influence.

Fish collection (lake trout, longnose suckers, round whitefish, lake chub) took place from the entire system between 2002 and 2015 using gill nets and electrofishing. Gill nets were generally set at three independent sites per lake. We used standardized nylon monofilament gill nets with mesh sizes increasing from 1.27 to 13.97 cm by 1.27 cm increments. We also sampled two locations per lake by electrofishing. Sampling always took place in the mid- to late summer during the short window of time when climate and weather allow access to these lakes. Sampled fish were measured for fork length and weight and were assessed for sex and maturity. Their otoliths (lake trout and round whitefish) and opercula (longnose suckers) were taken in situ for ageing. The ageing structures used are those accepted for the species aged; they have been validated for their accuracy and quantified for precision (see, e.g., Sotola et al. 2014; Campana et al. 2008). Otoliths work for all species; however, opercula will also work for suckers. Sucker opercula and otoliths have been compared and results were not significantly different ($\alpha = 0.05$). We used opercula for this species because they are far easier to remove from the fish than otoliths. Finclips (all four species) taken for DNA extraction were stored dry or in 95% ethanol. Number of individuals per species and lake are listed on Table 1. In total, considering all lakes, we collected and processed for DNA extraction 867 lake trout, 869 longnose suckers, 456 round whitefish, and 734 lake chub (Table 1). While lake trout were abundant in all sampled lakes, longnose sucker, round whitefish, and lake chub were absent or present only in small numbers in lakes Hawk and Genetics B. We did not collect lake chub from Cabot Lake, as this is a deep fjord lake with limited areas suitable for electrofishing. Details of the ageing procedure for longnose suckers based on opercula growth rings are available in Salisbury et al. (2016).

Life history analyses

Salisbury et al. (2016) estimated generation time, T, for longnose sucker as $T_{\text{LNS}} \approx 12$–$13.5$ years. In the present paper, we estimate T for lake trout (from Birch 1948) where

Fig. 1. Study area showing all lakes and connecting tributaries in the Kogaluk River catchment in northern Labrador (lakes were sampled for lake trout, longnose sucker, round whitefish, and lake chub). The Kogaluk River drains into the Atlantic Ocean via Voisey Bay. Waterfalls (WF1 to WF5) are indicated by small bars; their approximate heights and angles as determined by ground surveys (Anderson 1985) are as follows: WF1 ≈ 15 m and 90°, WF2 ≈ 12 m and 90°; WF3 ≈ 5.4 m and 90°; WF4 ≈ 5.4 m and 90°; WF5 ≈ 9 m and 60°–90°. Mistastin Lake was not included in the analysis. Lake trout were successfully collected from all lakes visited. No longnose suckers, round whitefish, or lake chub were collected from Genetics B or Hawk lakes. No lake chub were collected from Cabot Lake. Sample sizes per species and lake are available in Table 1. Figure modified from McCracken et al. (2013).



$$T = \frac{\sum x l_x m_x}{\sum l_x m_x}$$

This requires knowledge of l_x (the probability of survival to age x) and of m_x (age-specific fecundity or the number of offspring produced by an individual of age x). To estimate l_x (for each age class), we used the Robson–Chapman annual survivorship estimate ($\hat{S}$; Chapman and Robson 1960; Robson and Chapman 1961) using the age composition of gillnet-caught samples. We assumed $l_0 = 1$, $l_1 = \hat{S}^1 \times l_0$, … $l_\omega = \hat{S}^\omega \times l_0$, where ω is the maximum age observed from the sampled lake (Waples et al. 2014). To estimate m_x, we require knowledge of the age at 50% maturity (α) and of the adult life-span (AL). Age at 50% maturity was estimated using a binomial logistic regression (Harry et al. 2013) with age as the independent variable and maturity (0 as immature, 1 as mature) as the dependent variable in R (R Core Team 2013). AL was estimated as AL = $\omega - \alpha + 1$. Details of the analyses for longnose suckers are outlined in Salisbury et al. (2016). Fecundity for lake trout was estimated for each age class from α (rounded down to the nearest age) to ω and divided by 2 to approximate m_x (the number of offspring produced by an individual of age x) to account for the fact that only half of the population is female. For round whitefish, age at 50% maturity from similar high-latitude locations in Ungava Bay is $\alpha \approx 4+$ years (Armstrong et al. 1977; Morin et al. 1982). Generation time for round whitefish is therefore estimated at $T \approx 6.5$–7 years assuming age-specific survival rate among whitefish is similar to that of lake trout. For lake chub, age at 50% maturity was assumed to be $\alpha \approx 2$ years based on information in Bruce and Parsons (1976). Generation time for this small cyprinid species is therefore likely $3 < T < 4$.

DNA extraction, amplification, and genotyping

The final number of microsatellite loci genotyped for each species were 12 (lake trout), 17 (longnose sucker), 12 (round whitefish), and 19 (lake chub) (data to be made available in DRYAD). Below we describe the general procedures and note when these differ among the four species. Fin tissue samples (adipose fin for lake trout and round whitefish, pectoral or caudal for longnose sucker and lake chub) were digested with Proteinase K (Bio Basic Inc., Markham, Ontario, Canada) at 55 °C for ~8 h. DNA was then extracted from the resulting digest using a Glassmilk protocol modified from Elphinstone et al. (2003) with a Perkin Elmer Multiprobe II plus liquid handling system (Perkin Elmer, Waltham, Massachusetts). Random selections of DNA samples were electrophoresed on 1%–2% agarose gel and compared against a size standard to ensure sufficient quantity and quality of DNA for

Table 1. Summary statistics for nine lakes of the Kogaluk River catchment sampled between 2006 and 2015.

	N	H_O	H_E	A_r	$\hat{N}_e$	$\hat{N}_e$ (likely migrants removed)
Lake trout (*Salvelinus namaycush*)						
Lake 1	119 [119]	0.495	0.494	4.41	10 217 (215–∞)	10 217 (215–∞)
Genetics H	111 [110]	0.510	0.508	4.14	204 (93–4812)	203 (91–544)
Slushy	86 [84]	0.508	0.489	3.71	143 (63–4299)	141 (60–4299)
Strange	159 [159]	0.471	0.479	4.02	116 (78–196)	116 (78–196)
Esker–WP152	157 [155]	0.465	0.470	5.42	350 (133–∞)	323 (125–∞)
T-Bone	40 [40]	0.530	0.531	4.74	68 (32–433)	68 (32–433)
Cabot	80 [79]	0.565	0.576	4.94	630 (120–∞)	436 (110–∞)
Genetics B	49 [49]	0.500	0.466	4.29	50 (29–122)	50 (29–122)
Hawk	66 [66]	0.438	0.409	3.44	56 (27–211)	56 (27–211)
Total	867					
Longnose sucker (*Catostomus catostomus*)						
Lake 1	59 [57]	0.66	0.65	8.28	558 (202–∞)	356 (160–∞)
Genetics H	201 [194]	0.63	0.64	9.45	168 (135–217)	162 (130–210)
Slushy	103 [99]	0.65	0.64	9.92	314 (209–590)	314 (209–590)
Strange	122 [118]	0.61	0.61	8.48	821 (383–∞)	932 (402–∞)
Esker–WP152	212 [210]	0.66	0.66	14.09	2 740 (1 017–∞)	2 740 (1 017–∞)
T-Bone	115 [114]	0.63	0.64	9.62	NA (820–∞)	12 177 (744–∞)
Cabot	57 [56]	0.65	0.65	10.52	1 197 (302–∞)	1 197 (336–∞)
Total	869					
Round whitefish (*Prosopium cylindraceum*)						
Lake 1	18 [18]	0.414	0.430	2.59	86 (18–∞)	86 (18–∞)
Genetics H	87 [86]	0.487	0.473	2.73	507 (118–∞)	418 (108–∞)
Slushy	91 [91]	0.424	0.432	2.59	566 (98–∞)	566 (98–∞)
Strange	86 [84]	0.512	0.499	2.76	151 (78–639)	146 (76–590)
Esker–WP152	19 [19]	0.510	0.541	3.89	94 (33–∞)	94 (33–∞)
T-Bone	61 [59]	0.452	0.478	2.78	NA (131–∞)	NA (130–∞)
Cabot	94 [93]	0.441	0.434	2.65	234 (81–∞)	177 (75–∞)
Total	456					
Lake chub (*Couesius plumbeus*)						
Lake 1	95 [93]	0.429	0.438	3.81	454 (161–∞)	459 (163–∞)
Genetics H	111 [109]	0.487	0.493	3.8	471 (188–∞)	445 (185–∞)
Slushy	113 [112]	0.516	0.513	4.18	280 (156–923)	272 (151–907)
Strange	19 [18]	NA	NA	NA	NA (212–∞)	NA (257–∞)
Esker–WP152	304 [301]	0.518	0.529	7.82	1 040 (497–23 560)	987 (481–12 120)
T-Bone	92 [91]	0.498	0.501	3.94	815 (196–∞)	922 (199–∞)
Cabot	NA	NA	NA	NA	NA	NA
Total	734					

Note: Information provided: N = sample size [numbers in square brackets indicate sample size after removal of potential migrants]; H_O = observed heterozygosity; H_E = expected heterozygosity; A_r = allelic richness; $\hat{N}_e$ = estimate of effective population size. Lake trout were the only species sampled from Lakes Genetics B and Hawk. No lake chub were collected from Lake Cabot. Lake areas (km^2) are as follows: Lake 1, 11.3; Genetics H, 2.81; Slushy, 2.99; Strange, 2.09; Esker–WP152, 53.94; T-Bone, 19.76; Cabot, 25.39; Genetics B, 9.71; Hawk, 57.4.

subsequent polymerase chain reactions. Further details regarding the choice of microsatellite markers and related procedures for lake trout are available in McCracken et al. (2013), while details for longnose sucker are available in McCracken et al. (2014*a*) and Salisbury et al. (2016). Species-specific microsatellites for lake chub and round whitefish were chosen from McCracken et al. (2014*b* and 2014*c*, respectively) based on scoring ease and consistency.

Genetic quality control analyses

Individual genotypes were collected using SAGA Automated Microsatellite Software 3.3 (LI-COR Biosciences, Lincoln, Nebraska) followed by rigorous manual checking to ensure scoring accuracy. MICROCHECKER 2.2.3 (van Oosterhout et al. 2004) was used to test for the presence of null alleles or scoring inconsistencies.

Genetic analyses

Genotypic linkage and conformity to Hardy–Weinberg proportions as well as observed (H_O) and expected (H_E) heterozygosities were tested with Arlequin 3.5.1.2 (Excoffier and Lischer 2010). Linkage between pairs of loci were estimated using 10 000 permutations, while conformity to Hardy–Weinberg proportions was tested for each locus, population, and species using 1 000 000 steps in Markov chain and 100 000 dememorization steps. Results were then subject to false discovery rate correction (Benjamini and Hochberg 1995) to maintain an overall type 1 error probability at 0.05. Allele frequencies and allelic richness were estimated using FSTAT (Goudet 2001). Genetic differentiation (F_{ST}) was estimated with MSA 4.05 (Dieringer and Schlötterer 2003) using 100 000 individual permutations. These values were then linearized ($F_{ST}/(1 - F_{ST})$) following the procedure by Rousset (1997). Principal coordinates analyses were conducted using GenAlex 6.501 (Peakall and Smouse 2006).

Population structure analysis

Population structure was examined with the program STRUCTURE 2.3.4 (Hubisz et al. 2009), through principal coordinate analyses conducted on the matrices of linearized pairwise $\hat{F}_{ST}$ values using GENALEX and through a series of hierarchical AMOVAs conducted with Arlequin version 3.5 (Excoffier et al. 2007; Excoffier and Lischer 2010). STRUCTURE analyses were conducted

hierarchically for all four species, first examining the entire data set and identifying clusters, which were then independently subject to further STRUCTURE analyses. This process was continued on individual clusters until no further evidence of population structure was detected. We estimated the most likely number of clusters based on the Evanno methodology (Evanno et al. 2005) implemented in STRUCTURE HARVESTER version 0.6.92 (Earl et al. 2012). For all species, STRUCTURE runs were replicated 10 times at each level and K with each replicate run for 1 500 000 iterations with an initial burn-in of 500 000. Results of the separate replications for the most likely K were then combined into a single population output using the program CLUMPP 1.1.2 (Jakobsson and Rosenberg 2007) and visualized using the program DISTRUCT 1.1 (Rosenberg 2004). STRUCTURE runs for longnose sucker are described in Salisbury et al. (2016).

Identification of migrant individuals, effective population size (N_e), and gene flow estimation

Potential migrants were identified with GeneClass2 (Piry et al. 2004). N_e values were estimated both before and after the removal of individuals identified as potential migrants. Effective population size estimates ($\hat{N}_e$) were obtained with the linkage disequilibrium method implemented in LDNe (Waples and Do 2008). The program implements a bias correction for cases when the sample sizes are smaller than the actual effective population size (Waples 2006). Estimation via LDNe was conducted using the critical value for allele frequencies (P_{crit}) greater than 0.02, as described by Waples and Do (2010), as the vast majority of our sample sizes were >25, with 95% confidence intervals generated via jackknifing between pairs of loci. Estimates of effective population size for longnose sucker are those published in Salisbury et al. (2016) ; they were obtained by first estimating the effective number of breeders ($\hat{N}_b$) from a single cohort or two to three pooled cohorts when single cohort sample sizes were low. These estimates were then extrapolated to $\hat{N}_e$ using the empirical relationships described in Waples et al. (2013, 2014; see Salisbury et al. 2016). We were unable to follow the same procedure for the other species either because fish were not aged (lake chub) or because of uncertainty in the value of adult life-span (required for estimating N_e from N_b) and because the large number of age classes present, along with the fact that not all genotyped individuals were aged (lake trout, round whitefish), resulted in small cohort sizes.

Gene flow was estimated using BayesAss + (Wilson and Rannala 2003), which uses a Bayesian framework to infer recent migration rate. BayesAss + was run for 50 000 000 iterations with an initial burn in of 5 000 000, mixing parameters varied by species so as to achieve acceptance rates between 0.2 and 0.6.

Historical colonization assessment

We used DIYABC version 2.0 (Cornuet et al. 2014) to assess the likelihood of colonization from the west-southwest versus colonization from the east-southeast for lake trout and round whitefish, as well as for longnose sucker, the colonization route for which was originally described by Salisbury et al. (2016). As no lake chub samples were available for Cabot Lake, a key location for the testing of colonization from the west versus east, no analysis of potential colonization route using DIYABC was performed for this species. The approximate Bayesian computation (ABC) approach consists of generating simulated data sets under a given set of parameters (model), then selecting the simulated data sets that are closest to the observed data set and estimating the posterior distributions of the parameters (Cornuet et al. 2014). Different models can then be compared. The models in each simulated scenario describe how the sampled populations relate to a common ancestor under a given mutational model. Two extant populations can have a common ancestor, in which case, looking back in time, they merged a given number of generations in the past. Alternatively, a single extant population can be the result of admixture between two or more ancestral populations, in which case, looking back in time, the extant population was split between them (see Cornuet et al. 2014 for further details). Our first scenario depicted colonization from the west-southwest via the paleolake Naskaupi (Jansson and Kleman 2004). Under this scenario, T-Bone is the ancestral lake or the first lake to be colonized and from which fish expanded into the remaining lakes. The second scenario assumed colonization from the east-southeast, implying colonization via the coast (Black et al. 1986). For freshwater species, a scenario of colonization via the coast would in principle be possible through the lens of fresh water that would have been formed along the coast during periods of intense deglaciation. Under such a scenario, the first lake to be colonized would have been Cabot Lake (see Fig. 5B). Thus, the major difference between the two scenarios is whether colonization took place from the west-southwest (scenario 1) or the east-southeast (scenario 2). The lakes were immediately adjacent to Lake Nauskapi between 8400 and 7000 years BP (Jansson and Kleman 2004). Colonization was assumed to occur ca. 9000 years BP after the Laurentide Ice Sheet is thought to have retreated from the Kogaluk (Bryson et al. 1969; Short and Nichols 1977). Further colonization model details are described in the Results.

Generation times, T, were estimated to be 24, 12–13.5, 6.5–7, and 3–4 years for lake trout, longnose sucker, round whitefish, and lake chub, respectively (see Results and Salisbury et al. (2016) for longnose suckers). The prior distributions of time points differed across species, with maximum number of generations that elapsed since colonization set at 1000 for lake trout except for t_5 and t_7 (both time points involving Cabot Lake, which likely was in existence beforehand), which were set at 3000 for longnose sucker and 4000 for round whitefish.

In total, 6 000 000 simulations were run for each species. A Generalized Stepwise Mutation Model was assumed, and the prior for the mean mutation rate across loci was a uniform distribution between 1×10^{-4} and 1×10^{-3}, while individual locus mutation rates were allowed to range between 1×10^{-5} and 1×10^{-2}. The default value of the parameter P for the geometric distribution was used. N_e values were allowed to be variable among lakes, to range uniformly between 10 and 10 000, and were assumed to be constant through time. The one-sample summary statistics employed for generation of simulated data sets included mean number of alleles and mean size variance. Two-sample summary statistics included mean genic diversity, classification index, and shared allele distance.

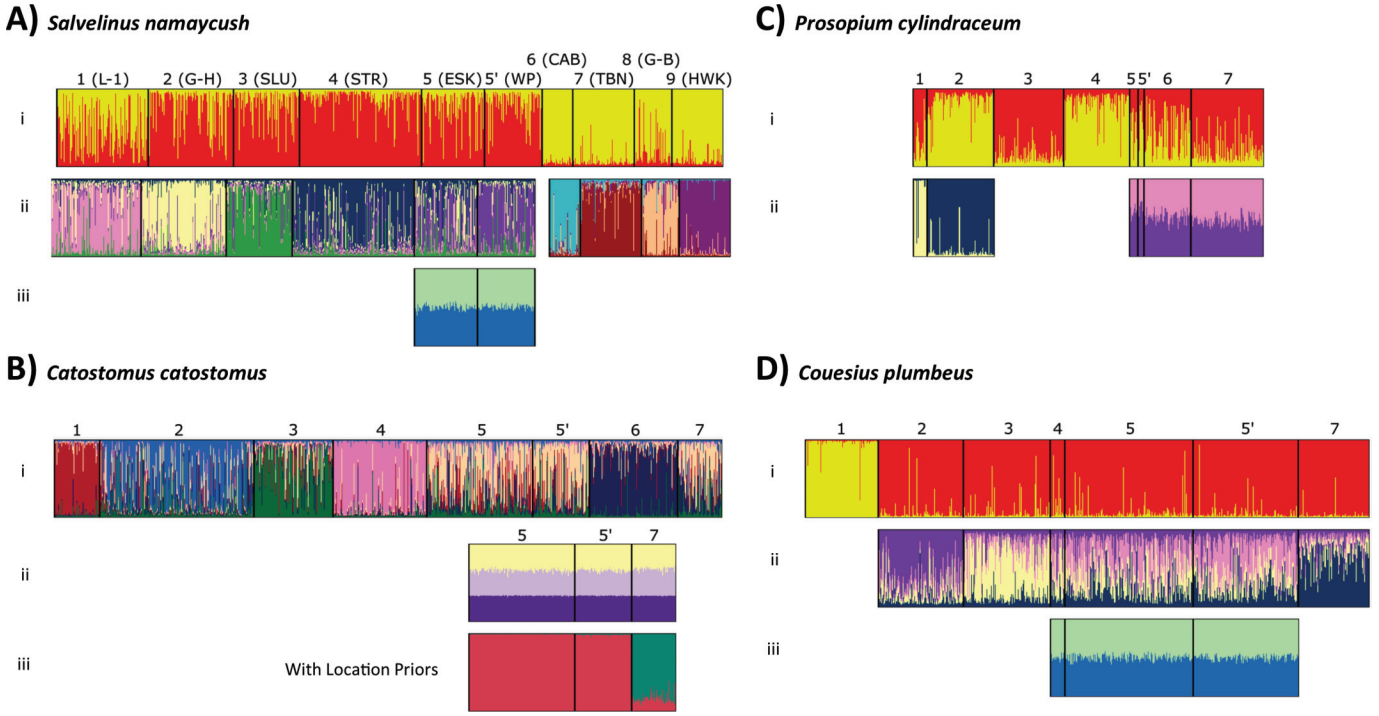
A principal components analysis was used to pre-evaluate the similarity between scenario-generated data sets and the observed data set. The posterior probabilities of both scenarios were assessed with the logistic regression method. Linear regression was used to determine the logit-transformed posterior parameters' distributions, using 1% of the closest simulated data sets. Bias and precision were estimated for each scenario using 500 pseudo-observed test data sets simulated using the original parameters from the 1% subset of the closest simulated data sets. Type I and type II error rates were generated for each scenario using confidence estimates derived from 500 pseudo-observed test data sets simulated using the original parameters. Model checking was completed for each scenario using five summary statistics not used in the initial data set generation as suggested by Cornuet et al. (2010): mean genic diversity (one-sample), the two-sample mean number of alleles, mean size variance, $\hat{F}_{ST}$, and $\delta\mu^2$ distance.

Results

Life history: generation time

For lake trout, age at 50% maturity (α) and generation time (T) were estimated as $\alpha \approx 15$ years and $T \approx 24$ years, respectively (refer

Fig. 2. Hierarchical population STRUCTURE analysis for (A) lake trout (*Salvelinus namaycush*) based on 12 microsatellite loci, (B) longnose sucker (*Catostomus catostotmus*) based on 17 microsatellite loci, (C) round whitefish (*Prosopium cylindraceum*) based on 12 microsatellite loci, and (D) lake chub (*Couesius plumbeus*) based on 19 microsatellite markers. Vertical coloured lines represent individual admixture coefficients. No differentiation was detected among fish from Lakes Esker and WP152 in any of the four species. Thus, individuals from Esker and WP152 were considered as a single population for all four species. [Colour online.]



on online Supplement 1[1]). Estimates for longnose suckers ($\alpha \approx$ 10 years and $T \approx$ 12–13.5 years) were taken from Salisbury et al. (2016) and those for round whitefish ($T \approx$ 6.5–7 years) and lake chub ($T \approx$ 3–4 years) were inferred from knowledge of α = 4+ years for round whitefish (Armstrong et al. 1977; Morin et al. 1982) and α = 2+ years for lake chub (Bruce and Parsons 1976). The four fish species therefore cover the range of plausible scenarios with $T \approx$ 4 years to $T \approx$ 24 years, suggesting that since deglaciation in northern Labrador around 9000 years BP (Bryson et al. 1969; Short and Nichols 1977), the number of generations elapsed may range from 3000 generations for lake chub to approximately 1500 for round whitefish, 900 for longnose sucker (Salisbury et al. 2016), and 500 for lake trout.

General statistics

Two previous studies on this system, one on lake trout (McCracken et al. 2013) and the other on longnose suckers (Salisbury et al. 2016), reported results based on $N \approx$ 560 and N = 869 individuals, respectively. Here, we report results based on N = 867 lake trout genotyped at 12 microsatellite markers, N = 869 longnose suckers genotyped at 17 microsatellite markers, N = 456 round whitefish genotyped at 12 microsatellite loci, and N = 734 lake chub genotyped at 19 microsatellites. Basic statistical data for longnose sucker were reported by Salisbury et al. (2016) and are presented here again for completeness (Table 1). The median value of missing data per locus and population, respectively, were 4.5% and 5.4% for lake trout, 2.1% and 2.2% for round whitefish, and 2.0% and 1.8% for lake chub. There was no consistent evidence across populations for departures from Hardy–Weinberg proportions for any locus and species or for linkage disequilibrium between pairs of loci for any of the species. All loci were therefore retained for subsequent analyses for all species. Observed and expected heterozygosities (H_O, H_E) as well as allelic richness (A_r) averaged over loci are reported in Table 1 (further details available in the online Supplement 1[1]).

Population structure, genetic diversity, and gene flow

Figure 2 depicts the STRUCTURE results for all four species. Lake trout was the only species collected from all nine lakes, while no longnose sucker, round whitefish, or lake chub were successfully collected from lakes Genetics B and Hawk, and no lake chub were collected from Cabot Lake either. All four species required at least a two-level hierarchical STRUCTURE analysis, with lake trout and lake chub also requiring a third level (Fig. 2). At the highest hierarchical level, lake trout clustered into two groups: a northern group comprising lake trout populations in Lake 1, Genetics H, Slushy, Strange, and Esker–WP152 and a southern group comprising Cabot Lake, T-Bone Lake, Genetics B, and Hawk (Fig. 2A-*i*). A level-2 hierarchical analysis, where the northern group was examined separately, revealed individual lake-level population structure with the exception of Lakes Esker and WP152. A level-3 hierarchical analysis revealed no difference between Esker and WP152 fish. Lake trout from Esker and WP152 were thus considered as belonging to a single population in all subsequent analyses.

For longnose sucker, population structure was largely revealed at the highest hierarchical level, with the exceptions of individuals from lakes Esker, WP152, and T-Bone (Fig. 2B-*i*). The level-2 analysis revealed no differences among these three populations (Fig. 2B-*ii*) unless the analysis was conducted with location priors, which distinguished suckers from T-Bone lake from those of Esker and WP152, which could not be distinguished from each other

[1]Supplementary data are available with the article through the journal Web site at http://nrcresearchpress.com/doi/suppl/10.1139/cjfas-2018-0416.

Table 2. Hierarchical analyses of molecular variance (AMOVAs) for each of the four species.

	Source of variation	df	SS	Variance components	% variation
LT, $K = 2$	(L-1, G-H, SLU, STR, ESK–WP) (TBN, CAB, G-B, HWK)				
	Among groups	1	65.49	0.079	7.62
	Among populations within groups	8	88.29	0.057	5.54
	Among individuals within populations	1764	1584.28	0.898	86.84
	Total	1773	1738.05	1.034	
LT, $K = 9$	(L-1) (G-H) (SLU) (STR) (ESK–WP) (TBN) (CAB) (G-B) (HWK)				
	Among groups	8	150.95	0.082	8.30
	Among populations within groups	1	2.83	0.012	1.19
	Among individuals within populations	1764	1584.28	0.898	90.50
	Total	1773	1738.05	0.992	
LT, $K = 7$	(L-1) (G-H) (SLU) (STR) (ESK–WP) (TBN) (CAB)				
	Among groups	6	95.88	0.060	6.05
	Among populations within groups	1	2.83	0.012	1.19
	Among individuals within populations	1528	1398.75	0.915	92.76
	Total	1535	1497.46	0.987	
LNS, $K = 7$	(L-1) (G-H) (SLU) (STR) (ESK–WP) (TBN) (CAB)				
	Among groups	6	72.69	0.049	6.72
	Among populations within groups	1	−0.31	0.002	−0.27
	Within populations	1730	1190.97	0.688	93.54
	Total	1737	1263.96	0.736	
RWF, $K = 5$, STRUCTURE result	(L-1) (G-H) (SLU) (STR) (ESK, WP, TBN, CAB)				
	Among groups	4	205.12	0.186	7.31
	Among populations within groups	3	32.41	0.128	5.01
	Among individuals within populations	904	2018.39	2.233	87.68
	Total	911	2255.93	2.547	
RWF, $K = 7$	(L-1) (G-H) (SLU) (STR) (ESK–WP) (TBN) (CAB)				
	Among groups	6	233.32	0.188	7.45
	Among populations within groups	1	4.21	0.107	4.22
	Among individuals within populations	904	2018.39	2.233	88.33
	Total	911	2255.93	2.528	
LCHUB, $K = 5$, STRUCTURE result	(L-1) (G-H) (SLU) (STR, ESK, WP) (TBN)				
	Among groups	4	236.40	0.181	6.26
	Among populations within groups	2	18.48	0.037	1.28
	Among individuals within populations	1461	3912.95	2.678	92.46
	Total	1467	4167.84	2.897	

Note: Species studied were lake trout (LT) under $K = 2$, $K = 9$, and $K = 7$, longnose sucker (LNS) under $K = 7$ (only ESK and WP152 pooled), round whitefish (RWF) under $K = 5$ (STRUCTURE results) and $K = 7$ (only ESK and WP152 pooled), and lake chub (LCHUB) under $K = 5$ (STRUCTURE results). df, degrees of freedom; SS, sum of squares; L-1, Lake 1; G-H, Genetics H; SLU, Slushy; STR, Strange; ESK–WP, Esker–WP152; TBN, T-bone; CAB, Cabot; G-B, Genetics B; HWK, Hawk.

(Fig. 2B-*iii*). Like lake trout, longnose suckers from these two lakes were thus also considered as belonging to a single population in all subsequent analyses.

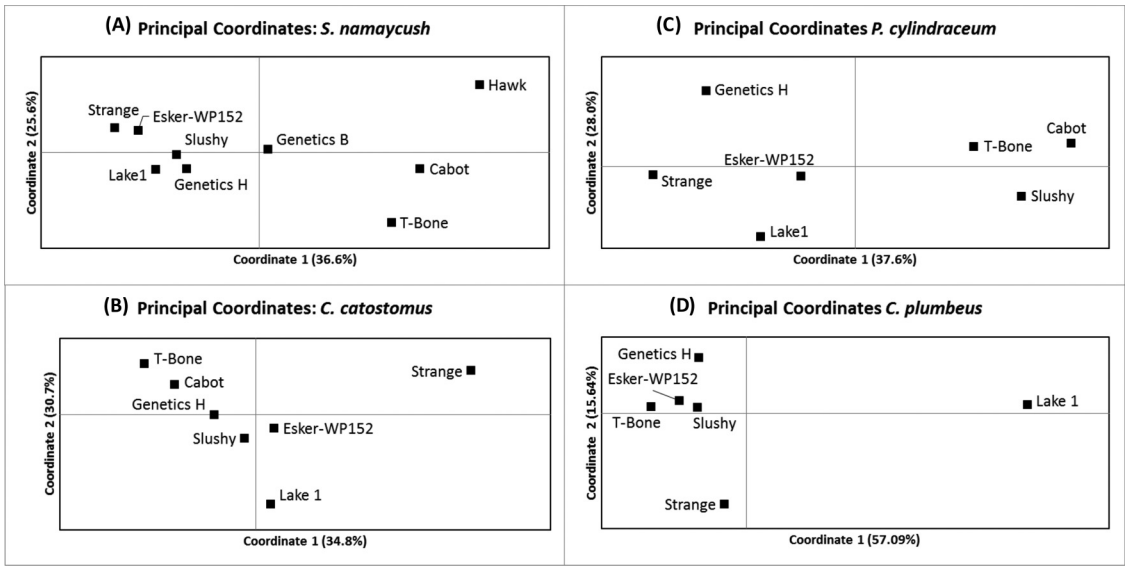
As was the case for longnose sucker above, round whitefish populations could also largely be distinguished at the highest hierarchical level, with the exception of individuals from Lakes Esker and WP152. This may be a consequence of the very few individuals collected from these two lakes. Similarly, for individuals from T-Bone and Cabot lakes (Fig. 2C), though they were somewhat distinguishable at the highest level, they could not be differentiated clearly in the level-2 analysis (Fig. 2C-*ii*). As for lake trout, location priors for whitefish did not affect the results.

Lastly, for lake chub, the highest hierarchical level involving all populations distinguishes individuals from Lake 1 from all other populations sampled. The second level analysis distinguishes the remaining populations with the exception, once again, of the individuals inhabiting Lakes Esker and WP152, as well as those from Lake Strange, where very few individuals were collected (Fig. 2D). Esker and WP152 lake chub were also considered as belonging to a single population in all subsequent analyses.

The AMOVA analyses conducted following the STRUCTURE results revealed very similar levels of population structure across all four species, with 6%–8% of the total variance explained by variation among groups. This is true in particular, for the three species (i.e., lake trout, longnose sucker, and round whitefish) present in at least eight lakes (seven after pooling Esker and WP152), where the AMOVAs using $K = 7$ with each individual lake, except Esker and WP152, forming a group; the percentages of variance explained by the groupings are 6.05% (lake trout), 6.72% (longnose sucker), and 7.45% (round whitefish). For these three species at least, therefore, the groupings are the same (see Table 2). (Notice that in these comparisons, collections from Genetics B and HAWK were removed since they are only available for lake trout.) For lake trout, the AMOVA involving all nine populations reveals the relatively large genetic differentiation that exists between the groups north and south of the Kogaluk River, with 7.62% of the total variation explained by differences between groups (Table 2). This percentage increases to 8.30% when most populations (except Esker and WP152) are considered individually (Table 2). For round whitefish, we conducted an extra AMOVA with $K = 5$ to mimic the STRUCTURE results where T-Bone and Cabot lakes are pooled with Esker and WP152 largely because of the small size of the Esker and WP152 samples (see Fig. 2C). The percentage of the total variation explained by differences among groups is slightly lower than that when $K = 7$ (7.31% versus 7.45%; Table 2).

Principal coordinate analysis (PCoA) based on pairwise linearized $\hat{F}_{ST}$ estimates largely reflect the species-specific STRUCTURE results (Fig. 3). For lake trout, a plot of the first two axes separate

Fig. 3. Principal coordinate analysis based on the matrix of pairwise linearized F_{ST} estimates for (A) lake trout (*S. namaycush*), (B) longnose sucker (*C. catostotmus*), (C) round whitefish (*P. cylindraceum*), and (D) lake chub (*C. plumbeus*). (Values in parentheses indicate the percentage of genetic variation explained by the principal coordinate.)



the southern from the northern group populations along axis 1 (36.6% variance explained; Fig. 3A). For longnose sucker, axis 1 (34.8% of variance) largely separates the population in Strange Lake from the rest (Fig. 3B). For round whitefish (Fig. 3C), axis 1 of the PCoA (37.6% of variance) separates populations in Lakes Slushy, Cabot, and T-Bone from the rest, and finally for lake chub, axis 1 (57.1% of variance; Fig. 3D) clearly distinguishes the population inhabiting Lake 1 from the rest.

Effective population size (SLU,STR,ESK$_e$)

N_e values were estimated with and without individuals identified as potential migrants for all four species (Table 1). For lake trout, only six potential migrants were identified over nine lakes with no individual population exhibiting fewer than two potential migrants. Removal of these individuals caused no significant change in $\hat{N}_e$ (Table 1; lake trout). Three lake trout populations exhibited $\hat{N}_e < 100$, and all three were in the southern group of populations (T-Bone, Genetics B, Hawk). Three more populations exhibited $100 < \hat{N}_e \leq 200$ (Genetics H, Slushy, and Strange). The estimates of effective population size for lake trout were relatively high in Cabot lake ($400 < \hat{N}_e \leq 600$) and in particular for those in Lake 1 ($\hat{N}_e \approx 10\,000$). For lake trout, median $\hat{N}_{e(\mathrm{LT,\ 9\ pops})} = 143$ over the nine populations sampled.

For longnose suckers, 21 potential migrants were identified across all seven lakes, with the population from Genetics H containing the highest number of potential migrants ($N = 7$) but also having the second largest sample size ($N = 201$). Regardless, as with the lake trout populations, removal of potential migrants had negligible effect on $\hat{N}_e$ for the longnose sucker populations, with the possible exception of Lake 1 where $\hat{N}_e$ declined from $\hat{N}_e = 558$ to $\hat{N}_e = 356$ with the removal of just two individuals (Table 1). In contrast with lake trout, there was no longnose sucker population with an $\hat{N}_e < 100$ and only one with $\hat{N}_e < 200$ (Genetics H $\hat{N}_e \approx$ 162–168; Table 1). For three of the seven sampled longnose sucker populations, $\hat{N}_e > 1000$. The median $\hat{N}_e$ for longnose suckers was higher than that for lake trout (median $\hat{N}_{e(\mathrm{LNS,\ 7\ pops})} = 689 >$ median $\hat{N}_{e(\mathrm{LT,\ 7\ pops})} = 204$; estimated over the seven common lakes).

For round whitefish, only six individuals overall were identified as potential migrants, and no individual population had fewer than two potential migrants removed. As for the previous two species, removal of potential migrants had negligible to no effect on $\hat{N}_e$ (Table 1). $\hat{N}_e < 100$ for two populations (Lake 1 and Esker–WP152), but sample sizes for both populations were small $N = 18$ and 19, respectively. Samples sizes for all other lakes were relatively high (i.e., $61 \leq N \leq 94$), and for these populations $150 < \hat{N}_e < 600$ with the exception of T-Bone where it could not be estimated. For round whitefish, the median $\hat{N}_{e(\mathrm{RWF,\ 6\ pops})} = 151$ considering six populations, and this value increased to median $\hat{N}_e = 370$ when the two populations with very small sample sizes were excluded.

For lake chub, we identified 11 potential migrants across the six lakes where this species was collected from, with the maximum number of potential migrants ($N = 3$) collected in Esker–WP152 where the sample size was largest ($N = 304$; Table 1). As with the previous species, the removal of potential migrants had little to no effect on $\hat{N}_e$. No lake chub population exhibited $\hat{N}_e < 100$, and the smallest $\hat{N}_e$ (i.e., $\hat{N}_e \approx 272$–280) was detected in Slushy lake; otherwise $445 \leq \hat{N}_e \leq 1040$ (Table 1), and the median $\hat{N}_{e(\mathrm{LCHB,\ 5\ pops})} =$ 471 among the five lakes where it could be estimated.

Overall, the median $\hat{N}_e$ among populations was lowest for lake trout (median $\hat{N}_{e(\mathrm{LT,\ 9\ and\ 7\ pops})} = 143$–204) followed by those for round whitefish (median $\hat{N}_{e(\mathrm{RWF,\ 4\ pops})} = 370$) and lake chub (median $\hat{N}_{e(\mathrm{LCHB,\ 5\ pops})} = 471$), with longnose sucker exhibiting the highest median $\hat{N}_e$ (median $\hat{N}_{e(\mathrm{LNS,\ 7\ pops})} = 689$). $\hat{N}_e$ and lake area were not correlated in any of the four species (data not shown). Overall, headwater populations tended to have larger $\hat{N}_e$ values than downstream populations. Interestingly, both lake trout and round whitefish, two of the three species collected in Cabot lake, the largest lake and the lake likely to receive migrants from all upstream lakes, exhibited $\hat{N}_e$ values that were lower than those exhibited by these species in some headwater lakes (Table 1).

Gene flow and dispersal direction

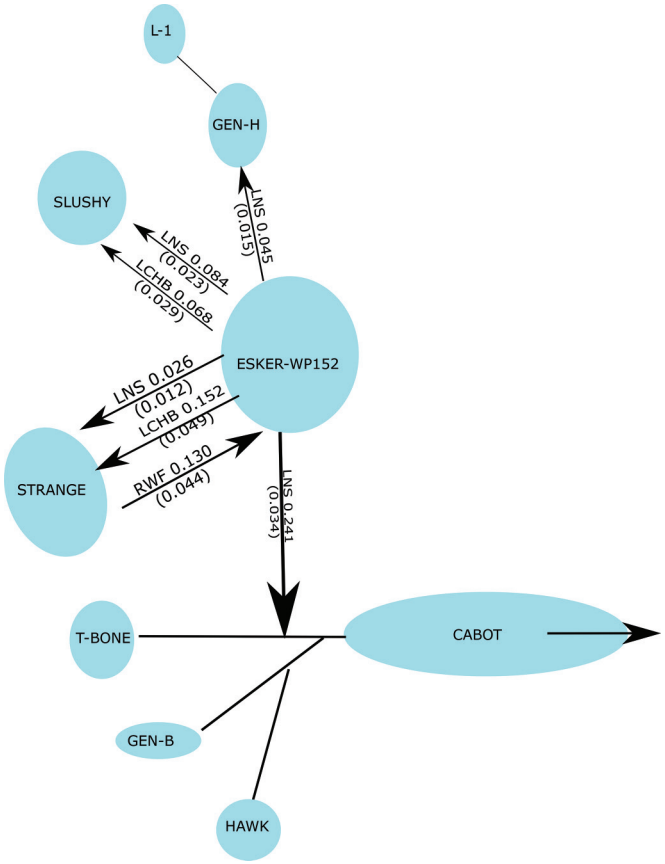
For lake trout, all estimates of gene flow m between lakes were nonsignificant with the possible exception of migration from Esker–WP152 to Genetics B (m (95% CI) = 0.091 (0.009–0.173); Table 3A). This is inconsistent with the fact that the lakes are located on opposite sides of the Kogaluk fjord. We therefore conclude there is no evidence that lake trout migrate between lakes in the Kogaluk River drainage. Unlike lake trout, longnose suckers from Lake Esker–WP152 exhibited gene flow > 0 to four other populations: low upstream gene flow from Esker–WP152 towards Genetics H, Slushy, and Strange and relatively high downstream gene flow to Cabot Lake (Table 3B). Gene flow between all other

Table 3. Gene flow estimates (migration rate, *m*, and 95% CI) obtained via BayesAss + 1.3.

	From								
Into	Lake 1	Genetics H	Slushy	Strange	Esker–WP152	T-bone	Cabot	Genetics B	Hawk
(A) Lake trout (*Salvelinus namaycush*)									
Lake 1	**0.939 (0.902–0.976)**	0.017 (–0.008–0.042)	0.011 (–0.007–0.029)	0.007 (–0.007–0.021)	0.008 (–0.006–0.022)	0.005 (–0.003–0.013)	0.004 (–0.004–0.012)	0.005 (–0.005–0.015)	0.004 (–0.004–0.012)
Genetics H	0.008 (–0.006–0.022)	**0.926 (0.887–0.965)**	0.006 (–0.006–0.018)	0.006 (–0006–0.018)	0.030 (–0.003–0.063)	0.003 (–0.003–0.009)	0.005 (–0.005–0.015)	0.005 (–0.005–0.015)	0.011 (–0.005–0.027)
Slushy	0.006 (–0.006–0.018)	0.008 (–0.006–0.022)	**0.912 (0.855–0.969)**	0.008 (–0.008–0.024)	0.048 (–0.003–0.099)	0.004 (–0.004–0.012)	0.005 (–0.003–0.013)	0.005 (–0.003–0.013)	0.005 (–0.005–0.015)
Strange	0.006 (–0.004–0.016)	0.006 (–0.004–0.016)	0.004 (–0.004–0.012)	**0.957 (0.924–0.990)**	0.017 (–0.010–0.044)	0.002 (0–0.004)	0.002 (–0.002–0.006)	0.003 (–0.001–0.007)	0.003 (–0.003–0.009)
Esker–WP152	0.008 (–0.006–0.022)	0.011 (–0.007–0.029)	0.006 (–0.006–0.018)	0.029 (–0.004–0.062)	**0.934 (0.891–0.977)**	0.002 (–0.002–0.006)	0.003 (–0.003–0.009)	0.003 (–0.003–0.009)	0.006 (–0.002–0.014)
T-bone	0.016 (–0.011–0.043)	0.010 (–0.010–0.030)	0.016 (–0.011–0.043)	0.012 (–0.012–0.036)	0.021 (–0012–0.054)	**0.874 (0.815–0.933)**	0.017 (–0.012–0.046)	0.021 (–0.010–0.052)	0.012 (–0.010–0.034)
Cabot	0.008 (–0.006–0.022)	0.006 (–0.004–0.016)	0.006 (–0.004–0.016)	0.005 (–0.005–0.015)	0.008 (–0.006–0.022)	0.005 (–0.005–0.015)	**0.948 (0.915–0.981)**	0.007 (–0.005–0.019)	0.008 (–0.006–0.022)
Genetics B	0.054 (–0.009–0.117)	0.011 (–0.011–0.033)	0.013 (–0.012–0.038)	0.053 (–0.014–0.120)	**0.091 (0.009–0.173)**	0.007 (–0.007–0.021)	0.011 (–0.011–0.033)	**0.741 (0.680–0.802)**	0.019 (–0.018–0.056)
Hawk	0.005 (–0.005–0.015)	0.005 (–0.005–0.015)	0.005 (–0.005–0.015)	0.007 (–0.007–0.021)	0.007 (–0.007–0.021)	0.005 (–0.005–0.015)	0.006 (–0.006–0.018)	0.005 (–0.005–0.015)	**0.954 (0.925–0.983)**
(B) Longnose sucker (*Catostomus catostomus*)									
Lake 1	**0.955 (0.924–0.986)**	0.006 (–0.006–0.018)	0.007 (–0.007–0.021)	0.007 (–0.007–0.021)	0.009 (–0.008–0.026)	0.010 (–0.007–0.027)	0.005 (–0.005–0.015)	—	–
Genetics H	0.005 (–0.003–0.013)	**0.913 (0.878–0.948)**	0.012 (–0.002–0.026)	0.007 (–0.003–0.017)	**0.045 (0.016–0.074)**	0.015 (–0.005–0.035)	0.002 (–0.002–0.006)	—	–
Slushy	0.008 (–0.006–0.022)	0.008 (–0.006–0.022)	**0.876 (0.829–0.923)**	0.008 (–0.006–0.022)	**0.084 (0.039–0.129)**	0.012 (–0.008–0.032)	0.004 (–0.004–0.012)	—	–
Strange	0.011 (–0.003–0.025)	0.003 (–0.003–0.009)	0.005 (–0.003–0.013)	**0.947 (0.922–0.972)**	**0.026 (0.002–0.050)**	0.005 (–0.005–0.015)	0.004 (–0.002– 0.010)	—	–
Esker–WP152	0.013 (–0.005–0.031)	0.015 (–0.003–0.033)	0.017 (–0.005–0.039)	0.011 (–0.003–0.025)	**0.935 (0.898–0.972)**	0.006 (–0.006–0.018)	0.003 (–0.003–0.009)	—	–
T-bone	0.006 (–0.004–0.016)	0.006 (–0.006–0.018)	0.005 (–0.005–0.015)	0.003 (–0.003–0.009)	0.009 (–0.007–0.025)	**0.967 (0.943–0.991)**	0.003 (–0.003–0.009)	—	–
Cabot	0.007 (–0.007–0.021)	0.008 (–0.006–0.022)	0.007 (–0.007–0.021)	0.009 (–0.007–0.025)	**0.241 (0.174–0.308)**	0.044 (–0.013–0.101)	**0.684 (0.659–0.709)**	—	–
(C) Round whitefish (*Prosopium cylindraceum*)									
Lake 1	**0.728 (0.638–0.818)**	0.015 (–0.014–0.044)	0.039 (–0.020–0.098)	0.107 (–0.007–0.221)	0.014 (–0.013–0.041)	0.030 (–0.023–0.083)	0.067 (–0.075–0.149)	—	–
Genetics H	0.004 (–0.004–0.012)	**0.959 (0.930–0.988)**	0.007 (–0.007–0.021)	0.008 (–0.008–0.024)	0.004 (–0.004–0.012)	0.009 (–0.007–0.025)	0.008 (–0.008–0.024)	—	–
Slushy	0.004 (–0.004–0.012)	0.005 (–0.005–0.015)	**0.967 (0.942–0.992)**	0.008 (–0.006–0.022)	0.004 (–0.004–0.012)	0.006 (–0.006–0.018)	0.007 (–0.005–0.019)	—	–
Strange	0.004 (–0.004–0.012)	0.010 (–0.008–0.028)	0.006 (–0.006–0.018)	**0.951 (0.916–0.986)**	0.005 (–0.005–0.015)	0.015 (–0.0.07–0.037)	0.011 (–0.007–0.029)	—	–
Esker–WP152	0.016 (–0.015–0.047)	0.015 (–0.012–0.042)	0.035 (–0.024–0.094)	**0.130 (0.044–0.216)**	**0.686 (0.649–0.723)**	0.033 (–0.028–0.094)	0.085 (–0.005–0.175)	—	–
T-bone	0.005 (–0.005–0.015)	0.010 (–0.008–0.028)	0.019 (–0.006–0.044)	0.006 (–0.006–0.018)	0.006 (–0.006–0.018)	**0.932 (0.885–0.979)**	0.021 (–0.014–0.056)	—	–
Cabot	0.004 (–0.004–0.012)	0.005 (–0.005–0.015)	0.012 (–0.010–0.034)	0.013 (–0.007–0.033)	0.004 (–0.004–0.012)	0.015 (–0.010–0.040)	**0.946 (0.907–0.985)**	—	–
(D) Lake chub (*Couesius plumbeus*)									
Lake 1	**0.974 (0.954–0.994)**	0.008 (–0.004–0.020)	0.005 (–0.003–0.013)	0.003 (–0.003–0.009)	0.006 (–0.004–0.016)	0.005 (–0.005–0.015)	—	—	–
Genetics H	0.004 (–0.004–0.012)	**0.962 (0.933–0.991)**	0.009 (–0.007–0.025)	0.003 (–0.003–0.009)	0.017 (–0.008–0.042)	0.006 (–0.006–0.018)	—	—	–
Slushy	0.005 (–0.005–0.015)	0.012 (–0.010–0.034)	**0.907 (0.846–0.968)**	0.003 (–0.003–0.009)	**0.068 (0.011–0.125)**	0.006 (–0.006–0.018)	—	—	–
Strange	0.010 (–0.010–0.030)	**0.096 (0.016–0.176)**	0.037 (–0.026–0.100)	**0.680 (0.654–0.705)**	**0.152 (0.056–0.248)**	0.025 (–0.024–0.074)	—	—	–
Esker–WP152	0.002 (–0.002–0.006)	0.004 (–0.004–0.012)	0.010 (–0.004–0.024)	0.001 (–0.001–0.003)	**0.977 (0.959–0.995)**	0.006 (–0.002–0.014)	—	—	–
T-bone	0.008 (–0.004–0.020)	0.008 (–0.006–0.022)	0.006 (–0.006–0.018)	0.003 (–0.003–0.009)	0.012 (–0.008–0.032)	**0.962 (0.933–0.991)**	—	—	–

Note: Columns are the source and rows are the recipient populations: (A) lake trout (generally little gene flow among lake trout populations in the Kogaluk River catchment); (B) longnose sucker; (C) round whitefish; and (D) lake chub. Significant (α = 0.05) estimates are indicated in bold.

Fig. 4. Schematic representation of gene flow estimates ($\hat{m}$) all species. Lake trout exhibited no gene flow among lakes. Longnose suckers exhibited upstream gene flow from Esker–WP152 into Slushy and Strange and downstream gene flow into Cabot Lake. Round whitefish exhibited downstream gene flow from Strange into Esker–WP152, and lake chub exhibited upstream gene flow from Esker–WP152 into Slushy and Strange. Numbers in parentheses indicate the standard deviation of the estimates.



longnose sucker populations was nonsignificant (Table 3B). Round whitefish exhibited downstream gene flow $m > 0$ from Strange Lake to Lake Esker–WP152 (m (95% CI) = 0.130 (0.044–0.216) Table 3C) and was otherwise nil. Finally, lake chub exhibited upstream gene flow $m > 0$ from Lake Esker–WP152 to Lakes Slushy and Strange, a pattern similar, at least partially, to that observed for longnose sucker (Table 3D and Fig. 4). Thus, all instances in which gene flow was distinctly larger than zero involved Lake Esker–WP152 mostly as a source of gene flow.

Colonization history (Fig. 5)

Figure 5 presents the two contrasting potential colonization scenarios and their posterior probabilities as a function of the stringency threshold used for three of the species examined in this study: (Fig. 5A) lake trout, (Fig. 5B) longnose sucker, and (Fig. 5C) round whitefish. For all three species, scenario 1 reflects colonization from the southwest via the proglacial Lake Nauskapi. Under this scenario (scenario 1), the ancestral population first colonized T-Bone Lake from which fish expanded into the remainder of the drainage. Scenario 2, instead, reflects colonization from the east via the sea for all three species. Under this scenario (scenario 2), the ancestral population first colonized Cabot Lake from which fish expanded into the remainder of the drainage. Potential colonization scenarios involve nine lakes (populations) for lake trout but only seven lakes for each of longnose sucker and round whitefish. For all three species, the relative posterior probability of scenario 1 is much higher than that of scenario 2 (Fig. 5).

The ancestral lake trout population first colonized T-Bone Lake from which lake trout expanded more or less simultaneously to Hawk and Cabot lakes followed by colonization of Genetics B, Slushy, and Strange lakes. Subsequently, lake trout from Lakes Slushy and Strange admixed to expand into Esker–WP152. Then lake trout from Esker–WP152 colonized Genetics H from which lake trout colonized Lake 1. Over all lakes, the colonization process likely started approximately 700 generation ago and was complete 250 generations ago (Fig. 5A). Under scenario 2, ancestral lake trout first colonized Cabot Lake from which they expanded into Hawk, Genetics B, and T-Bone at t_7, t_6, and t_5, respectively. The pattern of colonization for the remaining lakes is the same as under scenario 1. Scenario 1 received the highest support, suggesting lake trout likely colonized the Kogaluk system from the southwest (Fig. 5A).

The same two scenarios or hypotheses were tested with longnose sucker (Fig. 5B) and round whitefish (Fig. 5C). Under scenario 1, the ancestral longnose sucker population first colonized T-Bone. From this lake suckers expanded into Strange followed by the more or less simultaneous colonization of Cabot and Slushy and the admixture of Slushy and Strange giving rise to Esker–WP152. Under scenario 1, the system's colonization by longnose sucker likely started approximately 350 generations ago and was complete approximately 100 generations ago. Colonization by round whitefish likely started ~1500 generations ago and was complete ~900 generations ago (Fig. 5*c*). Parameter posterior distributions and model fits are shown in the online Supplement 2[1].

Discussion

We have shown that historical processes are important drivers of the observed genetic structure in extant fish populations inhabiting a spatially fragmented system that has been undisturbed so far. This is true for all three species in our study where this issue was examined regardless of the number of generations that have elapsed since their estimated arrival to the area. However, all four species exhibit a hierarchical pattern of population structure with similar diversity patterns despite differences in effective population size ($\hat{N}_e$), generation interval, and even contemporary migration rate estimates. Combined, these results suggest that contemporary landscape is important and influenced colonization history. The effects of colonization history are still observable in the population structure. The four species examined, lake trout, longnose sucker, round whitefish, and lake chub are widespread throughout Canada and have close relatives in many temperate regions of the Northern Hemisphere. Understanding how they behave in this particular system that is free of direct anthropogenic interference can provide a baseline against which one can evaluate how other systems undergoing anthropogenic perturbations are performing. Importantly, our results suggest that theoretical predictions based on equilibrium scenarios may not be a good baseline for comparison in the case of species inhabiting similar lake networks. Below we discuss the details of our findings and the implications for other spatially fragmented systems in northern latitudes.

Three of the four species included in the present study most likely colonized the Kogaluk River drainage from the southwest. This is the most likely colonization scenario suggested by the ABC analysis for the three species with sufficient geographic sampling coverage for the testing of alternate colonization hypotheses (i.e., lake trout, longnose sucker, and round whitefish). These analyses were conducted with DIYABC (Cornuet et al. 2014), a software package that may not be able to capture detailed demographic processes but can still capture major demographic changes (Cabrera and Palsbøll 2017). It is thus possible that our scenarios may not have captured the precise and detailed chronological

Fig. 5. Kogaluk drainage potential colonization scenarios and their posterior probabilities as a function of the stringency threshold used for three of the four species examined in this study: (A) lake trout (*Salvelinus namaycush*), (B) longnose sucker (*Catostomus catostomus*), and (C) round whitefish (*Prosopium cylindraceum*). For all three species, scenario 1 reflects colonization from the southwest via the glacial lake Nauskapi. Under scenario 1 the ancestral population first colonized T-Bone Lake (TBN) from which fish expanded into the remainder of the drainage. Scenario 2, instead, reflects colonization from the east via the sea for all three species. Under scenario 2 the ancestral population first colonized Cabot Lake from where fish expanded into the remainder of the drainage. The system comprises nine major lakes, from north to south: Lake 1 (L-1), Genetics H (G-H), Slushy (SLU), Strange (STR), Esker–WP152 (EKW), T-Bone (TBN), Cabot (CAB), Genetic B (G-B), and Hawk (HWK) (See Fig. 1). Lake trout were successfully collected from all nine lakes, but no longnose sucker or round whitefish were found in the two lakes south of the Kogaluk River (G-B and HWK). Potential colonization scenarios thus involved nine lakes for lake trout but only seven lakes for both longnose sucker and round whitefish. The *Y* axis reflects time into the past (in number of generations indicated for scenario 1) starting with the contemporary population at time $t_0 = 0$. Times are not shown at scale. For all three species the relative likelihood of scenario 1 is much higher than that of scenario 2. (A) Lake trout: The ancestral lake trout population first colonized TBN lake from which lake trout expanded more or less simultaneously to HWK and CAB lakes (median estimate: 768–731 generations BP; for details see ES-lake trout) followed by colonization of G-B (694 generations BP) and SLU and STR (684 generations BP). Subsequently lake trout from lakes SLU and STR admixed to expand into EKW (439 generations BP). Then lake trout from EKW colonized G-H (370 generations BP) from which lake trout colonized L-1 (246 generations BP). Under scenario 2, ancestral lake trout first colonized CAB lake from which they expanded into HWK, G-B, and TBN at t_7, t_6, and t_5, respectively. The pattern of colonization for the remaining lakes is the same as under scenario 1. (B) Longnose sucker and (C) round whitefish: Both scenarios are the same for both species. Under scenario 1, the ancestral longnose sucker population first colonized TBN from which suckers expanded into STR (336 generations BP), followed by the more or less simultaneous colonization of CAB and SLU and the admixture of SLU and STR, giving rise to EKW (159 generations GBP). EKW subsequently expanded to colonize G-H (145 generations BP) from which fish subsequently expanded into L-1 (108 generations BP). Under scenario 2, ancestral longnose suckers first colonized CAB lake from which they expanded independently into EKW and TBN at times t_6 and t_5, respectively. STR then is colonized from TBN at time t_4 and STR subsequently gives rise to SLU at time t_2 while EKW gives rise to G-H at time t_3 and L-1 is formed from G-H at time t_1. The pattern of colonization for round whitefish is the same as that for longnose sucker with the exception of the timing (in number of generations during which the various populations have been created under scenario 1). Parameter posterior distributions and model fits are shown in ES2. The three panels on the right show estimates of the posterior probability (*Y* axis) of scenarios 1 and 2 estimated with a logistic regression. The proportion of the scenario is the dependent variable, and the difference between the observed and simulated data set summary statistics are the independent variables. Ten estimates corresponding to the top 1% (60 000) simulated data sets are shown. [Colour online.]

sequence of how the northern lakes were colonized. However, both scenarios examined are similar in this regard: they largely only differ in the location of the most ancestral population(s), whether those inhabiting the southwestern-most lake, T-Bone Lake (scenario 1) or those inhabiting the easternmost lake and the lake closest to the Kogaluk River mouth, Cabot Lake. We are thus confident our models capture the essence of the two alternate colonization routes for this system and species considered. In fact our results are consistent with Black et al. (1986), who argued fish are most likely to have colonized Labrador following the last glacial retreat via overland route across Quebec rather than from the sea (see Michaud et al. 2010).

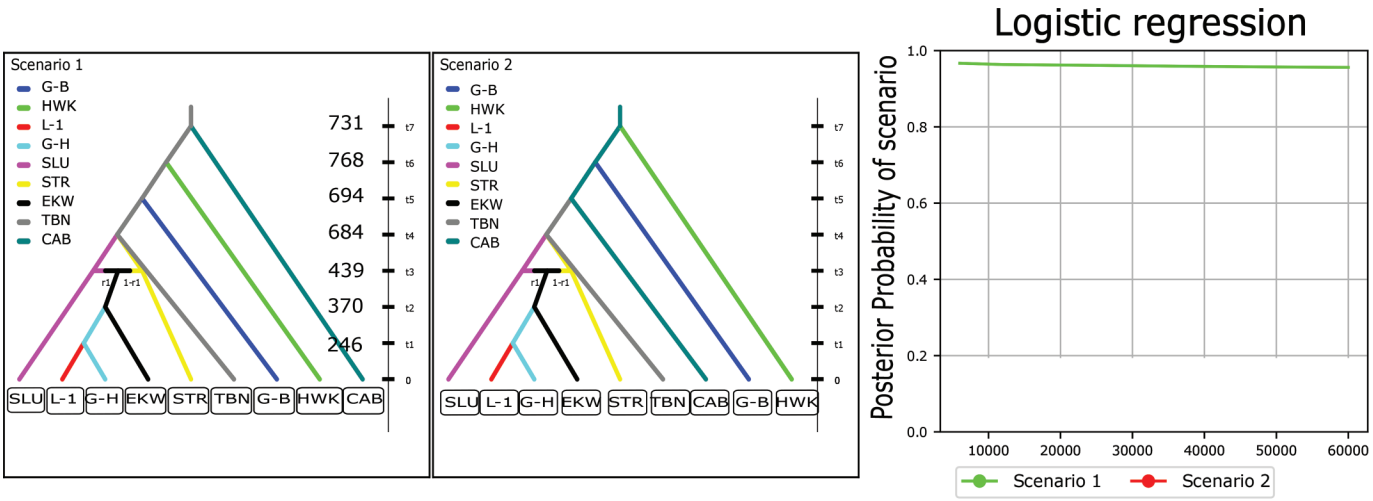
All four species exhibit a hierarchical pattern of contemporary population structure, with analyses consistently indicating that their respective Esker and WP152 populations are genetically indistinguishable from each other. Further, the percentage of the total genetic variation that is explained by differences among groups (where the groups ($K = 7$) are the same for at least three of the four species) is similar across species and ranges between 6% and 8%. Thus, all species exhibit similar diversity patterns despite the fact that they differ in $\hat{N}_e$ and generation interval, and at least three of them fit the same colonization model. Combined, these results suggest that landscape has driven the colonization history, and its effects can still be observed in the genetic structuring of all four species.

Gene flow (m) among populations was largely insignificant for lake trout and round whitefish. For longnose sucker though, gene flow was $m > 0$ out of the centrally located confluence Lake Esker–WP152, upstream to Slushy, Strange, and Genetics H and downstream to Cabot Lake. Similarly for lake chub, gene flow from Esker–WP152 was significant upstream to Slushy and Strange and downstream to Cabot Lake. Upstream migration for longnose sucker is consistent with the spawning migration for this species (Scott and Crossman 1998). Downstream migration for both longnose suckers and lake chub probably results from downstream passive transport of juveniles (Ryan 1980). Upstream migration is likely to result in an increase in the genetic diversity in headwater locations and in an erosion of their isolation (Salisbury et al. 2016). Overall, however, contemporary gene flow appears to play a relatively minor role in the distribution of genetic diversity for at least two of the species. One possibility is that colonization of downstream lakes was almost instantaneous, and this was followed by little migration. For these two species, therefore, lower diversity in downstream lakes may be due to a strong initial bottleneck and subsequent genetic drift affecting all lakes.

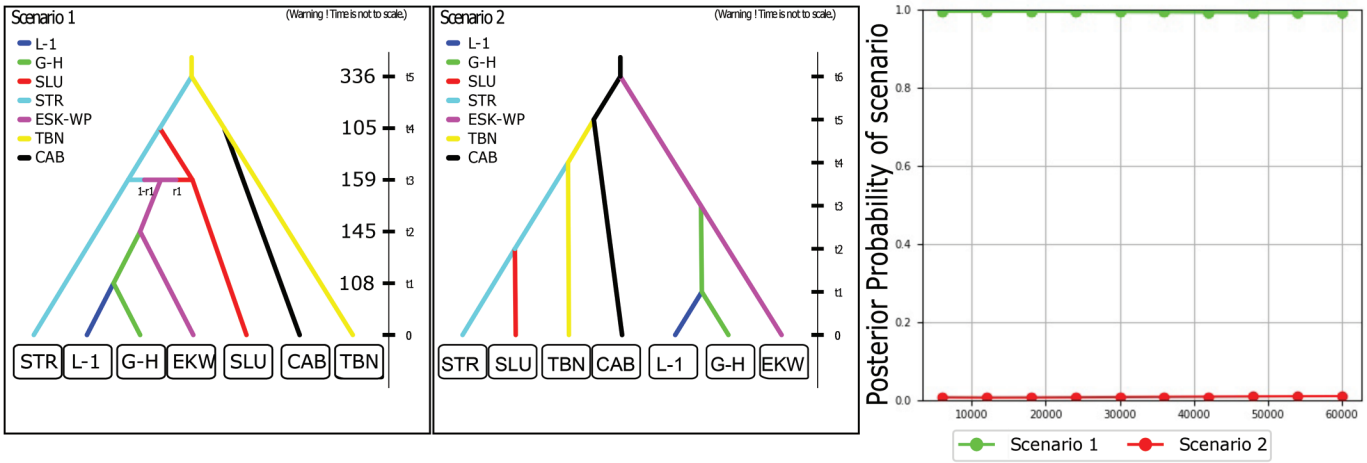
Fish collection took place over a period of 13 years from 2002 to 2015 (particularly for lake trout, longnose sucker, and round whitefish). Given the relatively long life-spans and generation intervals that characterize these species and the stable age structure composition typical of these remote, inaccessible, and unexploited subarctic lakes (Johnson 1976; Power 1978), it can reasonably be assumed that diversity and structure have not changed over this period.

Estimates of $\hat{N}_e$ for longnose sucker were obtained by first estimating the effective number of breeders (N_b) using single cohorts or pools of individuals of two to three cohorts when sample sizes for individual cohorts were too small. These were then extrapolated to $\hat{N}_e$ using the empirical relationships described by Waples and coworkers (Waples et al. 2013, 2014; Ruzzante et al. 2016; see Salisbury et al. 2016). Our ability to follow this procedure with the other two species for which age information was available for at least some genotyped individuals, lake trout and round whitefish, was limited because of the large number of age classes and cohorts in our samples resulting in low sample sizes per cohort and uncertainty in the value of the adult life-span, a value required for the application of Waples et al.'s (2013, 2014) empirical equations. This limitation notwithstanding, effective size estimates did not change significantly with the removal of potential migrants in any of the four species. Estimates were lowest for lake trout (median $\hat{N}_{e(LT, 9 \text{ and } 7 \text{ pops})}$ = 143–204) and highest for longnose sucker (median $\hat{N}_{e(LNS)}$ = 689), with those for round whitefish and lake chub being intermediate (Table 1). Species generally differed in the lakes in which their respective populations exhibited the largest $\hat{N}_e$. Lake trout exhibited the highest diversity in Lake 1, while

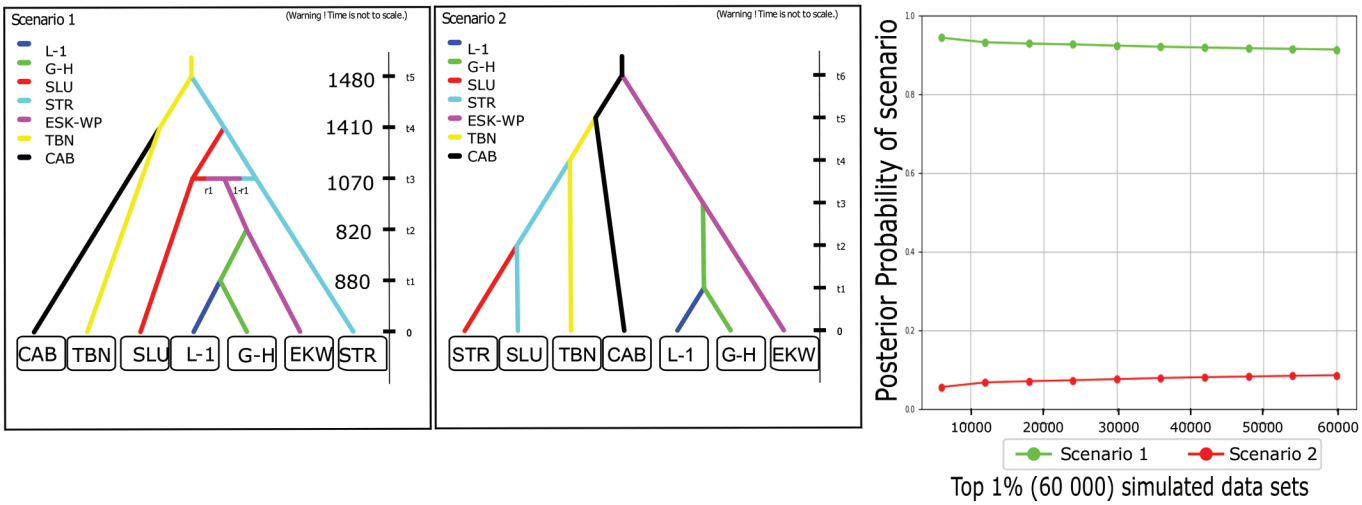
A) Lake trout (*Salvelinus namaycush)* colonization scenarios and relative likelihoods



B) Longnose sucker (*Catostomus catostomus)* colonization scenarios and relative likelihoods



C) Round whitefish (*Prosopium cylindraceum*) colonization scenarios and relative likelihoods



longnose sucker exhibited the greatest $\hat{N}_e$ in T-Bone Lake. Lake chub exhibited the largest $\hat{N}_e$ values in Esker–WP152 and T-Bone. Round whitefish, however, differed from the previous three species with its highest $\hat{N}_e$ values observed in populations inhabiting Lakes Slushy and Genetics H. Thus, though the four species differed in the lake in which they exhibited the largest $\hat{N}_e$, these were often found in headwater lakes.

Examination of the potential colonization scenarios requires knowledge of the number of generations that could have lapsed since deglaciation and hence knowledge of generation time. We estimated age at first maturity and generation time for lake trout as α = 15 years and T = 24 years, respectively. Our estimated generation time T therefore is longer than the value (T = 15) used by Harris et al. (2015) for lake trout inhabiting Great Bear Lake located 10° of latitude further north in the Northwest Territories than the Kogaluk River. Assuming these values have remained invariant through time, lake trout would have first colonized the southern lakes in the system (i.e., Cabot, Genetics B, Hawk) between ca. 10 500 and 16 800 BP. This is clearly unlikely or even unfeasible, since the area was covered by Laurentide Ice Sheet until ca. 9000 BP (Bryson et al. 1969; Short and Nichols 1977), and the Nauskapi Lake is thought to have first formed ca. 8000 BP (Jansson and Kleman 2004). However, neither age at first maturity nor generation time are likely to have remained invariant and may indeed have been much shorter than current values during the species' historical demographic expansion and colonization phase. Second, the fact that DIYABC has been shown not to be very accurate at recovering the timing of events (Cabrera and Palsbøll 2017) suggests caution should be exercised when interpreting the timing of demographic events. Disregarding issues of event timing, longnose sucker (α = 10 years, $T \approx 12$ years; Salisbury et al. 2016) would have first expanded and colonized the system between ca. 3400 and 4200 BP with the expansion–colonization probably complete by 1000–1200 BP. Similarly for round whitefish, with a generation time T = 6.5–7 years (Morin et al. 1982), colonization would have started a maximum of 9600–10 500 BP and would have been complete by as early as 5700–6200 BP. Assuming an age at first maturity $\alpha \approx 4$ years, the corresponding numbers would be 6000 and 3500 BP, respectively.

Regardless of the uncertainties and potential biases inherent with the ABC approach implemented in the DIYABC software package (Cabrera and Palsbøll 2017), the genetic variation observed still retains the footprint of the historical colonization pathways. This is true for all three species examined despite the differences in number of generations that are likely to have elapsed since their respective arrival to the region. One possibility is for the system to be in a transitional state where genetic diversity was originally seeded into the headwaters during colonization, with subsequent pooling of genetic diversity in downstream lakes such as Esker–WP152 and Cabot. However, the system as a whole is still out of migration–drift equilibrium because of its relatively young age and may at present only be transitioning to a dendritic system. We note though that the combination of upstream migration reported for two of the species along with the otherwise low migration rates observed and the long generation intervals described for several of the species would tend to slow down progress towards such migration–drift equilibrium (Salisbury et al. 2016). Landscape is also important. All species exhibit similar levels of contemporary population structure despite widely diverging generation times (sixfold variation from ~ 4 years for lake chub to perhaps as many as 24 years for lake trout) and effective sizes (range across species median $\hat{N}_e$ ~ 150 to ~700), both of which affect the rate of approach to equilibrium (Whitlock and McCauley 1999). These results suggest an important role for the landscape in influencing the observed structure (McCracken et al. 2013; Salisbury et al. 2016). First, it is likely that the streams connecting these lakes, which are generally frozen from October to May (Wheeler 1935) and exhibit reduced flow prior to the winter freeze (R. Perry, personal observation), offer little opportunity for substantial connectivity. Second, the various waterfalls in the system (Fig. 1) almost certainly prevent upstream migration. Thus, the relatively short time since the colonization event along with the limited gene flow among lakes, which delays a progression to equilibrium conditions, and the relatively long generation intervals have not allowed the species in the Kogaluk River drainage to reach an equilibrium between drift and gene flow. Our study clearly demonstrates that an improved understanding of historical and contemporary processes is required to fully explain observed patterns of structure and gene flow in metapopulation systems from regions affected by the Quaternary glaciations (see also Vera-Escalona et al. 2015, 2018). These studies highlight the importance of a temporal perspective on connectivity for the understanding of diversity in spatially complex metapopulations.

Data accessibility statement

Microsatellite genotypes will be submitted to DRYAD: doi: 10.5061/dryad.67n79r2.

Author contributions

DER, RP, and DK participated in the study design; RP, DK, DER, GRM, and SJS participated in the fieldwork; and GRM, SJS, and HTB produced the genetic data for the four species and together with DER and OEG were involved in the genetic analysis and interpretation. DER led the writing of the manuscript, with participation and input from all authors.

Acknowledgements

We thank Bill Green, Shawn Avery, Jerry Callahan, Shane Hann, Lorne Pike, and Reuben Solomon for their indispensable help collecting and processing samples in the field. We also thank the Newfoundland and Labrador Department of the Environment and Conservation for funding for fieldwork. Funds for the research were initially provided by an NSERC Northern Research Supplement Grant to DER followed by NSERC Strategic (STPGP 430198) and Discovery grants to DER. SJS is supported by Killam and NSERC scholarships. OEG was supported by the Marine Alliance for Science and Technology for Scotland (MASTS), which is funded by the Scottish Funding Council (grant reference HR09011).

References

Anderson, T.C. 1985. The rivers of Labrador. Can. Spec. Publ. Fish. Aquat. Sci. **81**: 1–389. Available from http://www.dfo-mpo.gc.ca/Library/89967.pdf.

Armstrong, J.W., Liston, C.R., Tack, P.I., and Anderson, R.C. 1977. Age, growth, maturity, and seasonal food habits of round Whitefish, *Prosopium cylindraceum*, in Lake Michigan near Ludington, Michigan. Trans. Am. Fish. Soc. **106**(2): 151–155. doi:10.1577/1548-8659(1977)106<151:AGMASF>2.0.CO;2.

Benjamini, Y., and Hochberg, Y. 1995. Controlling the false discovery rate: a practical and powerful approach to multiple testing. J. Royal Stat. Soc. Ser. B. **57**(1): 289–300. Available from https://www.jstor.org/stable/2346101.

Black, G.A., Dempson, J.B., and Bruce, W.J. 1986. Distribution and postglacial dispersal of freshwater fishes of Labrador. Can. J. Zool. **64**: 21–31. doi:10.1139/z86-005.

Bruce, W.J., and Parsons, R.F. 1976. Age, Growth and Maturity of Lake Chub (*Couesius plumbeus*) in Mile 66 Brook, Ten Mile Lake, Western Labrador. Fish. Mar. Serv. Res. Dev. Tech. Rep. Vol. 683.

Bryson, R.A., Wendland, W.M., Ives, J.D., and Andrews, J.T. 1969. Radiocarbon isochrones on the disintegration of the Laurentide Ice Sheet. Arct. Alp. Res. **1**: 1–13. doi:10.2307/1550356.

Cabrera, A.A., and Palsbøll, P.J. 2017. Inferring past demographic changes from contemporary genetic data: a simulation-based evaluation of the ABC methods implemented in DIYABC. Mol. Ecol. Resour. **17**: e94–e110. doi:10.1111/1755-0998.12696. PMID:28654208.

Caldera, E.J., and Bolnick, D.I. 2008. Effects of colonization history and landscape structure on genetic variation within and among threespine stickleback (*Gasterosteu aculeatus*) populations in a single catchment. Evol. Ecol. Res. **10**: 575–598. Available from http://www.evolutionary-ecology.com/issues/v10n04/hhar2292.pdf.

Campana, S.E., Casselman, J.M., and Jones, C.M. 2008. Bomb radiocarbon chronologies in the Arctic, with implications for the age validation of lake trout (*Salvelinus namaycush*) and other Arctic species. Can. J. Fish. Aquat. Sci. **65**: 733–743. doi:10.1139/f08-012.

Campbell, Grant, E.H., Lowe, W.H., and Fagan, W.F. 2007. Living in the branches:

population dynamics and ecological processes in dendritic networks. Ecol. Lett. **10**: 165–175. doi:10.1111/j.1461-0248.2006.01007.x. PMID:17257104.

Chapman, D.G., and Robson, D.S. 1960. The analysis of a catch curve. Biometrics, **16**: 354–368. doi:10.2307/2527687.

Cornuet, J.-M., Ravigné, V., and Estoup, A. 2010. Inference on population history and model checking using DNA sequence and microsatellite data with the software DIYABC (v1.0). BMC Bioinform. **11**: 401. doi:10.1186/1471-2105-11-401.

Cornuet, J.-M., Pudlo, P., Veyssier, J., Dehne-Garcia, A., Gautier, M., Leblois, R., et al. 2014. DIYABC v2.0: a software to make approximate Bayesian computation inferences about population history using single nucleotide polymorphism, DNA sequence and microsatellite data. Bioinformatics, **30**: 1187–1189. doi:10.1093/bioinformatics/btt763. PMID:24389659.

Dieringer, D., and Schlötterer, C. 2003. Microsatellite analyser (MSA): a platform independent analysis tool for large microsatellite data sets. Mol. Ecol. Notes, **3**(1): 167–169. doi:10.1046/j.1471-8286.2003.00351.x.

Earl, D.A., and von Holdt, B.M. 2012. STRUCTURE HARVESTER: a website and program for visualizing STRUCTURE output and implementing the Evanno method. Conserv. Genet. Resour. **4**: 359–361. doi:10.1007/s12686-011-9548-7.

Elphinstone, M.S., Hinten, G.N., Anderson, M.J., and Nock, C.J. 2003. An inexpensive and high-throughput procedure to extract and purify total genomic DNA for population studies. Mol. Ecol. Notes, **3**: 317–320. doi:10.1046/j.1471-8286.2003.00397.x.

Evanno, G., Regnaut, S., and Goudet, J. 2005. Detecting the number of clusters of individuals using the software structure: a simulation study. Mol. Ecol. **14**: 2611–2620. doi:10.1111/j.1365-294X.2005.02553.x. PMID:15969739.

Excoffier, L., and Lischer, H.E.L. 2010. Arlequin suite ver 3.5: a new series of programs to perform population genetics analyses under Linux and Windows. Mol. Ecol. Resour. **10**(3): 564–567. doi:10.1111/j.1755-0998.2010.02847.x. PMID:21565059.

Excoffier, L., Laval, G., and Schneider, S. 2007. Arlequin (version 3.0): an integrated software package for population genetics data analysis. Evol. Bioinform. Online, **1**: 47–50. PMID:19325852.

Fraser, C.I., Nikula, R., Ruzzante, D.E., and Waters, J.M. 2012. Poleward bound: biological impacts of Southern Hemisphere glaciations. Trends in Ecology and Evolution, **27**(8): 462–471.

Gomez-Uchida, D., Knight, T.W., and Ruzzante, D.E. 2009. Interaction of landscape and life history attributes on genetic diversity, neutral divergence and gene flow in a pristine community of salmonids. Mol. Ecol. **18**(23): 4857–4869. doi:10.1111/j.1365-294X.2009.04409.x.

Gomez-Uchida, D., Palstra, F.P., Knight, T.W., and Ruzzante, D.E. 2013. Contemporary effective population and metapopulation effective size (N_e and meta-N_e): Comparison among three salmonid species inhabiting a fragmented system and differing in gene flow and its asymmetries. Ecol. Evol. **3**(3): 569–580. doi:10.1002/ece3.485. PMID:23532448.

Goudet, J. 2001. FSTAT, a program to estimate and test gene diversities and fixation indices (version 2.9.3) [online]. Available from http://www.unil.ch/izea/softwares/fstat.html.

Harris, L.N., Chavarie, L., Bajno, R., Howland, K.L., Wiley, S.H., Tonn, W.M., and Taylor, E.B. 2015. Evolution and origin of sympatric shallow-water morphotypes of Lake Trout, *Salvelinus namaycush*, in Canada's Great Bear Lake. Heredity, **114**: 94–106. doi:10.1038/hdy.2014.74. PMID:25204304.

Harry, A.V., Tobin, A.J., and Simpfendorfer, C.A. 2013. Age, growth and reproductive biology of the spot-tail shark, *Carcharhinus sorrah*, and the Australian blacktip shark, *C. tilstoni*, from the Great Barrier Reef World Heritage Area, north-eastern Australia. Mar. Freshw. Res. **64**(4): 277–293. doi:10.1071/MF12142.

Hewitt, G. 2000. The genetic legacy of the Quaternary ice ages. Nature, **405**: 907–913.

Hössjer, O., Olsson, F., Laikre, L., and Ryman, N. 2014. A new general analytical approach for modeling patterns of genetic differentiation and effective size of subdivided populations over time. Math. Biosci. **258**: 113–133. doi:10.1016/j.mbs.2014.10.001. PMID:25445736.

Hössjer, O., Olsson, F., Laikre, L., and Ryman, N. 2015. Metapopulation inbreeding dynamics, effective size and subpopulation differentiation — a general analytical approach for diploid organisms. Theor. Popul. Biol. **102**: 40–59. doi:10.1016/j.tpb.2015.03.006. PMID:25875853.

Hubisz, M.J., Falush, D., Stephens, M., and Pritchard, J.K. 2009. Inferring weak population structure with the assistance of sample group information. Mol. Ecol. Resour. **9**: 1322–1332. doi:10.1111/j.1755-0998.2009.02591.x. PMID:21564903.

Jakobsson, M., and Rosenberg, N.A. 2007. CLUMPP: a cluster matching and permutation program for dealing with label switching and multimodality in analysis of population structure. Bioinformatics, **23**: 1801–1806. doi:10.1093/bioinformatics/btm233. PMID:17485429.

Jansson, K.N., and Kleman. J. 2004. Early Holocene glacial lake meltwater injections into the Labrador Sea and Ungava Bay. Paleoceanography, **19**(1). doi:10.1029/2003PA000943.

Johnson, L. 1976. Ecology of Arctic populations of lake trout, *Salvelinus namaycush*, lake whitefish, *Coregonus clupeaformis*, Arctic char, *S. alpinus* and associated species in unexploited lakes of the Canadian Northwest Territories. J. Fish. Res. Board Can. **33**(11): 2459–2488. doi:10.1139/f76-293.

Junker, J., Peter, A., Wagner, C.E., Mwaiko, S., Germann, B., Seehausen, O., and Keller, I. 2012. River fragmentation increases localized population genetic structure and enhances asymmetry of dispersal in bullhead (*Cottus gobio*). Conservation Genetics, **13**(2): 545–556. doi:10.1007/s10592-011-0306-x.

Labonne, J., Ravigné, V., Parisi, B., and Gaucherel, C. 2008. Linking dendritic network structures to population demogenetics: the downside of connectivity. Oikos, **117**: 1479–1490. doi:10.1111/j.0030-1299.2008.16976.x.

Manel, S., and Holderegger, R. 2013. Ten years of landscape genetics. Trends in Ecology and Evolution, **28**(10): 614–621.

Manel, S., Schwartz, M.K., Luikart, G., and Taberlet, P. 2003. Landscape genetics: combining landscape ecology and population genetics. Trends in Ecology and Evolution, **18**: 189–197.

McCracken, G.R., Perry, R., Keefe, D., and Ruzzante, D.E. 2013. Hierarchical population structure and diversity in a dendritic lake trout (*Salvelinus namaycush*) system in northern Labrador. Freshw. Biol. **58**(9): 1903–1917. doi:10.1111/fwb.12179.

McCracken, G.R., Brewis, H.T., McBride, M.C., Perry, R., Keefe, D., and Ruzzante, D.E. 2014*a*. Development and characterization of 36 novel microsatellite markers for lake chub (*Couesius plumbeus*) using Illumina paired-end sequencing. Conserv. Genet. Resour. **6**(4): 873–876. doi:10.1007/s12686-014-0228-2.

McCracken, G.R., Wilson, K.L., Paterson, I., Perry, R., Keefe, D., and Ruzzante, D.E. 2014*b* Development of 17 novel microsatellite markers for the longnose sucker (*Catostomus catostomus*) and successful cross-specific amplification of 14 previously developed markers from congeneric species. Conserv. Genet. Resour. **6**(2): 329–332. doi:10.1007/s12686-013-0086-3.

McCracken, G.R., Wilson, K.L., Brewis, H.T., McBride, M.C., Paterson, I., Perry, R., et al. 2014*c*. Development of 26 novel microsatellite makers for the round whitefish (*Prosopium cylindraceum*) and successful polymorphic cross-specific amplification from 7 previously developed salmonid markers. Conserv. Genet. Resour. **6**(4): 1023–1026. doi:10.1007/s12686-014-0276-7.

Michaud, W.K., Perry, R.C., Dempson, J.B., Shears, M., and Power, M. 2010. Occurrence of Lake Chub, *Couesius plumbeus*, in Northern Labrador. Can. Field Nat. **124**: 113–117. doi:10.22621/cfn.v124i2.1048.

Morin, R., Dodson, J.J., and Power, G. 1982. Life history variations of anadromous cisco (*Coregonus artedii)*, lake whitefish (*C. clupeaformes)*, and round whitefish (*Prosopiurn cylindraceum*) populations of eastern James–Hudson Bay. Can. J. Fish. Aquat. Sci. **39**(7): 958–947. doi:10.1139/f82-131.

Morrissey, M., and de Kerckhove, D.T. 2009. The maintenance of genetic variation due to asymmetric gene flow in dendritic metapopulations. Am. Nat. **174**: 875–889. doi:10.1086/648311. PMID:19860541.

Peakall, R., and Smouse, P.E. 2006. GENALEX 6: genetic analysis in Excel. Population genetic software for teaching and research. Mol. Ecol. Notes, **6**(1): 288–295. doi:10.1111/j.1471-8286.2005.01155.x.

Perkin, J.S., and Gido, K.B. 2012. Fragmentation alters stream fish community structure in dendritic ecological networks. Ecol. Appl. **22**(8): 2176–2187.

Piry, S., Alapetite, A., Cornuet, J.M., Paetkau, D., Baudouin, L., and Estoup, A. 2004. GeneClass2: a software for genetic assignment and first-generation migrant detection. J. Hered. **95**: 536–539. doi:10.1093/jhered/esh074. PMID:15475402.

Power, G. 1978. Fish population structure in Arctic lakes. J. Fish. Res. Board Can. **35**: 53–59. doi:10.1139/f78-008.

R Core Team. 2013 R: a language and environment for statistical computing [online]. R Foundation for Statistical Computing, Vienna, Austria. Available from http://www.R-project.org/.

Robson, D.S., and Chapman, D.G. 1961. Catch curves and mortality rates. Trans. Am. Fish. Soc. **90**: 181–189. doi:10.1577/1548-8659(1961)90[181:CCAMR]2.0.CO;2.

Rosenberg, N.A. 2004. DISTRUCT: a program for the graphical display of population structure. Mol. Ecol. Notes, **4**(1): 137–138. doi:10.1046/j.1471-8286.2003.00566.x.

Rousset, F. 1997. Genetic differentiation and estimation of gene flow from *F*-statistics under isolation-by-distance. Genetics, **145**: 1219–1228. PMID:9093870.

Ruzzante, D.E., Walde, S.J., Gosse, J.C., Cussac, V.E., Habit, E., Zemlak, T.S., and Adams, E.D.M. 2008. Climate control on ancestral population dynamics: insight from Patagonian fish phylogeography. Mol. Ecol. **17**: 2234–2244.

Ruzzante, D.E., McCracken, G.R, Parmelee, S., Hill, K., Corrigan, A., MacMillan, J., and Walde, S.J. 2016. Effective number of breeders, effective population size and their relationship with census size in an iteroparous species, *Salvelinus fontinalis*. Proc. Royal Soc. B. **283**(1823). doi:10.1098/rspb.2015.2601.

Ryan, P.M. 1980. Fishes of the Lower Churchill River, Labrador. Fisheries and Marine Service. Technical Report No. 922.

Salisbury, S.J., McCracken, G.R., Keefe, D., Perry, R., and Ruzzante, D.E. 2016. A portrait of a sucker: the landscape genetics of longnose sucker (*Catostomus catostomus*) in a spatially fragmented system. Mol. Ecol. **25**: 4126–4145. doi:10.1111/mec.13757. PMID:27393723.

Scott, W.B., and Crossman, E.J. 1998. Freshwater Fishes of Canada. Galt House Publications Ltd., Ontario, Canada.

Short, S.K., and Nichols, H. 1977. Holocene pollen diagrams from subarctic Labrador–Ungava: Vegetational history and climatic change. Arct. Alp. Res. **9**: 265–290. doi:10.2307/1550543.

Sotola, V.A., Maynard, G.A., Hayes-Pontius, E.M., Mihuc, T.B., Malchoff, M.H., and Marsden, J.E. 2014. Precision and bias of using opercles as compared to otoliths, dorsal spines, and scales to estimate ages of largemouth and smallmouth bass. Northeast. Nat. **21**: 565–573. doi:10.1656/045.021.0411.

Storfer, A., Murphy, M.A., Spear, S.F., Holderegger, R., and Waits, L.P. 2010. Landscape genetics: Where are we now? Mol. Ecol. **19**(17), 3496–3514.

Tufto, J., and Hindar, K. 2003. Effective size in management and conservation of subdivided populations. J. Theor. Biol. **222**: 273–281.

van Oosterhout, C., Hutchinson, W.F., Wills, D.P.M., and Shipley, P. 2004. MICRO-CHECKER: software for identifying and correcting genotyping errors in microsatellite data. Mol. Ecol. Notes, **4**: 535–538. doi:10.1111/j.1471-8286.2004.00684.x.

Vera-Escalona, I., Habit, E., and Ruzzante, D.E. 2015. Echoes of a distant time: effects of historical processes on contemporary genetic patterns in *Galaxias platei* in Patagonia. Mol. Ecol. **24**: 4112–4128. doi:10.1111/mec.13303. PMID:26147523.

Vera-Escalona, I., Senthivasan, S., Habit, E., and Ruzzante, D.E. 2018. Past, present, and future of a freshwater fish metapopulation in a threatened landscape. Conserv. Biol. **32**(4): 849–859. doi:10.1111/cobi.13093. PMID:29431232.

Waples, R.S. 2006. A bias correction for estimates of effective population size based on linkage disequilibrium at unlinked gene loci. Conserv. Genet. **7**: 167–184. doi:10.1007/s10592-005-9100-y.

Waples, R.S., and Do, C. 2008. LDNE: a program for estimating effective population size from data on linkage disequilibrium. Mol. Ecol. Resour. **8**: 753–756. doi:10.1111/j.1755-0998.2007.02061.x. PMID:21585883.

Waples, R.S., and Do, C. 2010. Linkage disequilibrium estimates of contemporary N_e using highly variable genetic markers: a largely untapped resources for applied conservation and evolution. Evol. Appl. **3**: 244–262. doi:10.1111/j.1752-4571.2009.00104.x. PMID:25567922.

Waples, R.S., Luikart, G., Faulkner, J.R., and Tallmon, D.A. 2013. Simple life history traits explain key effective population size ratios across diverse taxa. Proc. R. Soc. B. **280**: 20131339. doi:10.1098/rspb.2013.1339. PMID:23926150.

Waples, R.S., Antao, T., and Luikart, G. 2014. Effects of overlapping generations on linkage disequilibrium estimates of effective population size. Genetics, **197**: 769–780. doi:10.1534/genetics.114.164822. PMID:24717176.

Wheeler, E.P., 2nd. 1935. The Nain–Okak Section of Labrador. Geog. Rev. **25**(2): 240–254. doi:10.2307/209599.

Whitlock, M.C., and McCauley, D.E. 1999. Indirect measures of gene flow and migration: $F_{ST} \neq 1/(4Nm + 1)$. Heredity, **82**(2): 117–125. doi:10.1038/sj.hdy.6884960. PMID:10098262.

Wilson, G.A., and Rannala, B. 2003. Bayesian inference of recent migration rates using multilocus genotypes. Genetics, **163**: 1177–1191. PMID:12663554.